

What the First Stage Cannot Detect

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Abstract

Applied instrumental-variable (IV) practice reports a first-stage F , increasingly the conditional F of Sanderson and Windmeijer (2016), and reads a large value as a license to interpret the second-stage estimate. We show that no first-stage diagnostic can provide that license. We also show that a strong first stage misleads in two distinct ways. In just-identified linear IV with linear covariates, an exact decomposition writes the 2SLS bias as a covariance between instrument–propensity curvature and the covariate level functions. That same object enters the first-stage covariance the conditional F is built from. Under instrument–covariate collinearity the identifying signal vanishes and the F is inflated by this contamination, so the strength is manufactured (mode a). But even with an honest, enormous F the bias can hide on the outcome side, where no first-stage number reaches it (mode b). Detecting it requires the outcome. We give a directed conditional-moment test, built from reduced-form regressions alone. It has power exactly where the F is silent, and in its screening form its size is controlled even under weak identification, where a saturate-and-compare contrast collapses. In a panel of canonical designs it flags the 401(k) eligibility instrument. There an $F > 7,000$ and a contamination share $\varrho = 0.03$ are both reassuring, yet the linear-covariate estimate understates the wealth effect by about a third. The same test clears designs with harmless curvature on which a Ramsey RESET false-alarms.

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1. Introduction

The instrumental-variable (IV) estimator is the workhorse of applied microeconomics whenever treatment is suspected to be endogenous. Its credibility rests on two pillars, the exclusion restriction and the strength of the first stage. The first pillar is fundamentally untestable and is defended by argument and by sensitivity analysis (Conley et al., 2012; van Kippersluis and Rietveld, 2018). The second pillar is treated as a statistical object to be certified. A large literature, including Bound et al. (1995), Staiger and Stock (1997), Stock and Yogo (2005), Montiel Olea and Pflueger (2013), and Andrews et al. (2019), has produced a battery of rule-of-thumb and size-based procedures for diagnosing weak instruments. It is now routine, indeed mandatory in most journals, to report a first-stage F statistic. When the model contains multiple endogenous regressors or a rich set of controls, Sanderson and Windmeijer (2016) show that the relevant object is the *conditional* first-stage F , computed after partialling the covariates out of the excluded instrument. Applied work increasingly reports it. In the just-identified, single-instrument model we study, this conditional F coincides with the ordinary first-stage F on the covariate-residualized instrument. We retain the “conditional F ” label throughout. It is what applied work now reports, and the overidentified generalization (Section 9) is where the distinction bites.

Running in parallel, a second literature has clarified *what* a valid, strong instrument identifies under treatment-effect heterogeneity. Since Imbens and Angrist (1994) and Angrist et al. (1996), it has been understood that IV recovers a local average treatment effect (LATE), the effect on compliers, not the population average treatment effect. Heckman and Vytlacil (2005) formalize the role of essential heterogeneity. de Chaisemartin (2017) shows that failures of monotonicity contaminate the estimand with negatively weighted defier effects. Mogstad and Torgovitsky (2018) and Mogstad and Torgovitsky (2024) synthesize the modern understanding of IV under unobserved heterogeneity. Most consequential for empirical practice is Blandhol et al. (2022). They establish that once covariates enter the second stage *linearly*, which is the near-universal specification, 2SLS is in general *not* a convex (non-

negatively weighted) average of LATEs, unless the covariates are entered nonparametrically (“saturated”). Related work has sharpened this. Słoczyński (2022) treats the just-identified case. Goldsmith-Pinkham et al. (2024) give a general treatment of the “contamination bias” that additive covariate adjustment induces across *multiple* treatments. And Zhao et al. (2025) show that consistent estimation of interacted 2SLS requires the product of the IV propensity score and the covariates to be linear in the covariates.² It is by now well understood, then, that linearity of the instrument propensity $\mathbb{E}[Z \mid X]$ in the covariates is the pivotal condition for a LATE interpretation. Zhao et al. (2025) establish this condition as necessary and sufficient for the consistency of interacted 2SLS, and Blandhol et al. (2022) recommend assessing it directly with a Ramsey RESET test of the linearity of $\mathbb{E}[Z \mid X]$, a diagnostic on (Z, X) alone. Słoczyński et al. (2026) synthesize these diagnostics and the double/debiased-machine-learning alternatives that control for covariates flexibly.

Two questions this literature leaves open motivate the present paper, and they are the questions a practitioner actually faces. First, *what does the strength diagnostic tell you about all this?* The strength literature certifies that the instrument *moves the treatment*, and the heterogeneity literature characterizes *what the movement identifies*. The implicit bridge in applied work, that “a strong (conditional) first stage licenses a causal reading of the second stage,” has never been examined. We show it is false, and, more sharply, that *no* first-stage diagnostic can repair it. Second, *is checking the propensity enough?* We show it is not. A propensity-only check such as a RESET on $\mathbb{E}[Z \mid X]$ cannot distinguish a biased design from an unbiased one with equally nonlinear propensity, because whether the nonlinearity matters depends on the outcome. These two negative results, together with the constructive tests and characterizations that answer them, are what is new here. In independent concurrent work,

²We use “contamination” descriptively, for the object $\text{Cov}(h, m_D)$, $\text{Cov}(h, m_Y)$ in (3). It is related to but distinct from the contamination bias of Goldsmith-Pinkham et al. (2024), which arises from *multiple* treatments under additive adjustment and persists however flexibly the covariates are specified. Our object is single-treatment and vanishes when the instrument propensity is linear in X . A further, unrelated use of the term appears in Dossani et al. (2024), whose “contaminated controls” are endogenous covariates correlated with the instrument rather than a functional-form phenomenon.

Krumme and Westphal (2026) reach a related conclusion, that a diagnostic must be directed at the outcome rather than the propensity alone. We differ by isolating the contamination as a single conditional-moment restriction. It conditions on validity, remains valid under weak identification, and ties the bias to the first-stage strength itself. We defer the full contrast to the end of this section. These results build on a by-now-standard understanding that propensity linearity is what a linear covariate specification silently assumes.

This paper. We study just-identified linear IV with a binary instrument, a binary treatment, and covariates entered linearly, under conditional-on- X validity and monotonicity. Our starting point is an exact Frisch–Waugh–Lovell (FWL) decomposition of the 2SLS estimand in Lemma 1.

$$\beta_{2SLS} = \frac{\overbrace{\mathbb{E}[e(X)\{1 - e(X)\} p(X) \text{LATE}(X)]}^{\text{signal}} + \overbrace{\text{Cov}(h(X), m_Y(X))}^{\text{contamination}}}{\underbrace{\mathbb{E}[e(X)\{1 - e(X)\} p(X)]}_{\text{signal}} + \underbrace{\text{Cov}(h(X), m_D(X))}_{\text{contamination}}},$$

where $e(x) = \mathbb{P}(Z = 1 \mid X = x)$ is the instrument propensity, $h(x)$ is the component of $e(x)$ orthogonal to the linear covariate span (the “propensity curvature”), $p(x)$ is the conditional first stage, and m_D, m_Y are the covariate level functions. The signal terms are a convex-weighted average of conditional LATEs. Each contamination term is a covariance, so it vanishes exactly when the propensity curvature is orthogonal to the relevant level function, i.e. $\theta_D = 0 \iff h \perp m_D$ and $\theta_Y = 0 \iff h \perp m_Y$. Two sufficient conditions make both vanish at once. A propensity linear in X ($h \equiv 0$) reproduces Blandhol et al. (2022); level functions linear in X do the same. Orthogonality can also hold with neither function linear, so these are sufficient, not necessary.

We build on this decomposition. The signal ratio is the saturated-specification estimand of Blandhol et al. (2022), and the contamination term is the single-treatment, functional-form analogue of Goldsmith-Pinkham et al. (2024). But our subject is a feature of it that has gone unremarked. The *denominator* $S_D + \theta_D$ is exactly the first-stage covariance the conditional

F is built from, so the same nuisance that biases the estimand inflates the reported strength. Every result below turns on that identity.

Because the contamination sits in both the numerator and the denominator of the estimand, it produces *two distinct failure modes*, and separating them organizes the paper. Consider first **mode (a)**, *manufactured strength*, in which the instrument is collinear with the covariates. The denominator contamination θ_D dominates, the identifying signal S_D vanishes, and the conditional F is inflated by the very curvature that biases the estimand. Here the contamination share ϱ is large. Consider next **mode (b)**, *invisible outcome-side bias*, in which the strength is honest. Now θ_D and hence ϱ are small and the F is genuinely large, yet the estimand is biased through the numerator term θ_Y . That term loads on the outcome, and no first-stage number can see it. The two modes share one root cause. Strength is a first-stage object while the bias is an outcome object. The flagship 401(k) design turns out to be a clean instance of mode (b), not (a). Mode (a) is a limiting regime. The exact collinearity limit $\text{Var}(Z \mid X) \rightarrow 0$ is an idealization, and none of the canonical designs in Section 7 sits at it. Its relevance is that the approach to the limit is already consequential. Near-collinearity is common wherever the instrument is residualized against saturated fixed effects or a rich control set. Table 1 shows the estimand leaving the LATE hull at moderate collinearity, while $\mathbb{E}[e(1 - e)]$ is still around one quarter, far from zero, well before the limit is reached. Mode (b), by contrast, requires no collinearity at all and is the empirically operative case throughout our panel. Our results trace both. *First*, we characterize the role of instrument–covariate collinearity, the pure form of mode (a). Proposition 2 considers any sequence of designs in which Z becomes a deterministic function of X , the collinearity limit in which $\text{Var}(Z \mid X) \rightarrow 0$. Along such a sequence the signal terms vanish while the contamination terms do not. The estimand then converges to $\text{Cov}(h, m_Y) / \text{Cov}(h, m_D)$, a ratio of covariate contrasts with no causal content, which loses the convex-LATE interpretation and can fall outside the hull of $\{\text{LATE}(x)\}$ (as in our Monte Carlo), though whether it does is design-dependent.

Second, Corollary 1 shows the Sanderson–Windmeijer conditional first-stage slope stays bounded away from zero along the same sequence, so the conditional F , which scales with the sample size, *diverges* in n rather than collapsing as a weak-instrument diagnostic should. The reported strength is manufactured by the contamination, so a large conditional F places no bound on the bias. We sharpen this into an impossibility result (Proposition 3). *No* diagnostic computed from the first stage, meaning any functional of the joint law of (Z, D, X) and the conditional F in particular, can detect the contamination bias. The reason is that the bias depends on the conditional distribution of the *outcome*, which relevance diagnostics never use.

Third, we propose a *directed* conditional-moment test of the null of no contamination. The contamination terms $\theta_D = \mathbb{E}[h(X)D]$ and $\theta_Y = \mathbb{E}[h(X)Y]$ are functionals of observables that require only reduced-form regressions of Z , D and Y on X . The test therefore involves no endogenous object and is immune to the weak-identification problem that disables the natural alternative. That alternative estimates a saturated second stage and forms a Hausman contrast against an estimator that is itself weakly identified in exactly this regime. We give a strongly identified screening version, which tests the sufficient condition $\theta_D = \theta_Y = 0$ with a limiting χ^2_2 distribution and a consistent pairs bootstrap. We also give a bias-targeting version, which tests the necessary and sufficient restriction $\mathbb{E}[h(X)\{Y - \bar{L}D\}] = 0$. In a three-way comparison, the bias-targeting test is the only one of the three that both attains full power against genuine contamination and correctly certifies a design with *harmless* propensity curvature, one in which the instrument is nonlinear in the covariates but the estimand is unbiased. A Ramsey RESET on the first stage, being blind to the outcome, necessarily false-alarms in that design.

Fourth, we turn the same identity into a number a practitioner can report and a trade-off that disciplines its use. Define the *contamination share* ϱ . It is the fraction of the reported first-stage covariance, equivalently of the apparent conditional strength, that is due to propensity curvature rather than to complier variation. It is estimated from reduced-

form regressions, carries a bootstrap confidence interval, and is meant to be read next to the conditional F . A large F paired with a non-negligible ϱ signals that the strength is partly manufactured. The bias-corrected estimator that removes the contamination is itself standard. It is the residualized-propensity (saturated-covariate) IV estimator of Blandhol et al. (2022), so what is new is its *cost*. Corollary 2 makes the trade-off exact. The corrected estimator’s own first stage *is* the vanishing signal term, so the identifying variation it surrenders equals precisely the share ϱ that measured the manufactured strength (Proposition 8). Under collinearity one cannot both decontaminate and remain strongly identified. This no-free-lunch trade-off is why ϱ belongs in the first-stage report rather than in a footnote to it.

Our test is a member of the class of consistent conditional-moment specification tests (Newey, 1985; Tauchen, 1985; Bierens, 1990), and it is related in spirit to Ramsey’s RESET (Ramsey, 1969) and to overidentification and difference-in-J (“ C ”) tests. It differs in its *target*. Rather than testing linearity of a conditional mean in all directions, it probes the single direction that governs the contamination bias, namely the propensity curvature interacted with the outcome and treatment. It does so without estimating any IV coefficient. This is what lets it have power exactly where strength diagnostics are silent. As noted above, the closest antecedent is the concurrent Krumme and Westphal (2026). Their procedure extends testable-inequality implications for IV validity to a joint null of instrument validity and correct specification, whereas ours conditions on validity, isolates the contamination term, and remains valid under weak identification.

Section 2 sets up the model and proves the decomposition. Section 3 develops the collinearity limit and the failure of the conditional F . Section 4 constructs the test. Section 5 presents the bias-corrected estimator and the identification trade-off. Section 6 reports Monte Carlo evidence and Section 7 applies the diagnostic to a panel of canonical designs. Section 8 gives recommendations for practice, and Section 9 concludes. Proofs are collected in the Appendix.

2. Framework and the strength–bias identity

Let $Z \in \{0, 1\}$ be a binary instrument, $D \in \{0, 1\}$ a binary treatment, $Y \in \mathbb{R}$ an outcome, and $X \in \mathbb{R}^{d_x}$ a vector of covariates. Potential treatments are $D(0), D(1)$ and potential outcomes $Y(d, z)$. The observed variables are $D = D(Z)$ and $Y = Y(D, Z)$. We maintain the standard conditional-on- X LATE assumptions.

Assumption 1 (Conditional validity). $\{Y(d, z), D(z)\}_{d,z} \perp\!\!\!\perp Z \mid X$, and $Y(d, 1) = Y(d, 0) \equiv Y(d)$ (exclusion), almost surely.

Assumption 2 (Conditional monotonicity). $D_i(1) \geq D_i(0)$ almost surely.

Assumption 3 (Conditional relevance). $p(x) \equiv \mathbb{E}[D \mid Z = 1, X = x] - \mathbb{E}[D \mid Z = 0, X = x] > 0$ on a set of x of positive probability with instrument overlap $0 < e(x) < 1$. This overlap qualification is what makes the signal $S_D = \mathbb{E}[e(X)\{1 - e(X)\}p(X)]$ strictly positive, so that the convex-LATE benchmark $\bar{L} = S_Y/S_D$ is well defined. Relevance confined to strata where $e(x) \in \{0, 1\}$ would leave $S_D = 0$, and the estimand, though still defined through $\mathbb{E}[\tilde{Z}D] = \theta_D$ whenever $\theta_D \neq 0$, would be pure contamination with no $\bar{L}$ to benchmark against.

Assumption 4 (Maintained linear specification). The analyst estimates the just-identified linear IV model $Y = \beta D + X'\gamma + \varepsilon$, using Z as the single excluded instrument and $(1, X')$ as included instruments.

Assumption 5 (Regularity). Y, D, Z and X have finite second moments, and $\mathbb{E}[(1, X')'(1, X')]$ is nonsingular. In addition, $\mathbb{E}[\tilde{Z}D] \neq 0$, where $\tilde{Z} \equiv Z - L(Z \mid 1, X)$ and $L(\cdot \mid 1, X)$ denotes the population linear projection on $(1, X')$.

Under Assumption 4, the population 2SLS coefficient on D admits the familiar FWL / partialling-out form

$$\beta_{2SLS} = \frac{\mathbb{E}[\tilde{Z}Y]}{\mathbb{E}[\tilde{Z}D]}. \quad (1)$$

Define the instrument propensity $e(x) = \mathbb{P}(Z = 1 \mid X = x) = \mathbb{E}[Z \mid X = x]$, its linear projection $\ell(x) = L(Z \mid 1, X)(x)$, and the *propensity curvature*

$$h(x) \equiv e(x) - \ell(x), \quad \mathbb{E}[h(X)] = 0, \quad \mathbb{E}[h(X)r(X)] = 0 \text{ for all affine } r. \quad (2)$$

Let $m_D(x) = \mathbb{E}[D \mid X = x]$, $m_Y(x) = \mathbb{E}[Y \mid X = x]$, and let $\text{LATE}(x) = \mathbb{E}[Y(1) - Y(0) \mid D(1) > D(0), X = x]$ denote the conditional LATE.

Lemma 1 (Contamination decomposition). *Under Assumptions 1–5,*

$$\beta_{2SLS} = \frac{S_Y + \theta_Y}{S_D + \theta_D}, \quad (3)$$

where

$$S_Y = \mathbb{E}[e(X)\{1 - e(X)\}p(X)\text{LATE}(X)], \quad S_D = \mathbb{E}[e(X)\{1 - e(X)\}p(X)],$$

$$\theta_Y = \text{Cov}(h(X), m_Y(X)) = \mathbb{E}[h(X)Y], \quad \theta_D = \text{Cov}(h(X), m_D(X)) = \mathbb{E}[h(X)D].$$

Consequently, writing $\bar{L} \equiv S_Y/S_D$ for the convex-weighted average of conditional LATEs with weights proportional to $e(x)\{1 - e(x)\}p(x)$,

$$\beta_{2SLS} - \bar{L} = \frac{\theta_Y - \bar{L}\theta_D}{S_D + \theta_D}, \quad \text{with } S_D + \theta_D = \mathbb{E}[\tilde{Z}D]. \quad (4)$$

The signal terms S_D, S_Y are exactly the objects delivered by the saturated specification of Blandhol et al. (2022), and $S_Y/S_D = \bar{L}$ is a proper, non-negatively weighted average of conditional LATEs. The contamination terms θ_D, θ_Y are covariances between the propensity curvature h and the covariate level functions. By (2) they equal the covariance between the *nonlinear* part of the propensity and the *nonlinear* part of the level functions. As covariances they vanish exactly under orthogonality, $\theta_D = 0 \iff h \perp m_D$ and $\theta_Y = 0 \iff h \perp m_Y$, which a linear propensity ($h \equiv 0$) or a linear m_D (resp. m_Y) delivers as a sufficient special

case but which can also hold with neither linear. Linearity of a single level function kills only its own term. A linear propensity kills both and reproduces Blandhol et al. (2022). The signal ratio $\bar{L}$ is their saturated-specification estimand, and we take it as given. Our subject is a feature of the decomposition that has gone unremarked, which we state as the paper’s organizing result.

Proposition 1 (Strength–bias identity). *The denominator of the 2SLS estimand is the population first-stage covariance $\mathbb{E}[\tilde{Z}D] = S_D + \theta_D$ from which the Sanderson–Windmeijer conditional F is constructed. The contamination θ_D is the common term. It enters the denominator of the bias $\beta_{2SLS} - \bar{L} = (\theta_Y - \bar{L}\theta_D)/(S_D + \theta_D)$ and, being a signed covariance, inflates the reported first-stage covariance when $\theta_D > 0$ and deflates it when $\theta_D < 0$. The displacement of β_{2SLS} from $\bar{L}$ requires the numerator $\theta_Y - \bar{L}\theta_D \neq 0$, so strength and interpretability need not move together. The point of the identity is the weaker but consequential one that they can be driven by the same nuisance θ_D , and under collinearity (Section 3) they are.*

Proof. This is immediate from Lemma 1. The denominator of (3) is $S_D + \theta_D$, and $\mathbb{E}[\tilde{Z}D] = \mathbb{E}[(h(X) + (Z - e(X)))D] = \theta_D + S_D$ since $\mathbb{E}[(Z - e(X))D \mid X] = e(X)\{1 - e(X)\}p(X)$ integrates to S_D . $\square$

The identity organizes the two failure modes. Through the *denominator* the contamination inflates the reported strength. Under instrument–covariate collinearity this is the dominant effect and the strength is manufactured (mode a, Section 3). Through the *numerator*, via θ_Y , it biases the estimand even when the denominator, the reported strength, is honest. That bias loads entirely on the outcome (mode b, the 401(k) design of Section 7.1).

3. Collinearity and the failure of the first-stage diagnostic

We now formalize what happens as the instrument becomes collinear with the covariates. Intuitively, “ Z is collinear with X ” means Z is close to a (possibly nonlinear) function of X ,

i.e. the conditional variance $\text{Var}(Z \mid X) = e(X)\{1 - e(X)\}$ is small. We index a sequence of data-generating processes (DGPs) by t .

Assumption 6 (Collinearity sequence). *Along $t = 1, 2, \dots$ the following hold. (i) $e_t(x)\{1 - e_t(x)\} \rightarrow 0$ for almost every x and $\mathbb{E}[e_t(X)\{1 - e_t(X)\}] \rightarrow 0$. (ii) The propensity curvature converges, $h_t \rightarrow h^*$ in L^2 , with h^* non-degenerate. (iii) The conditional LATE, the conditional first stage, and the level functions converge in L^2 to limits $\text{LATE}^*, p^*, m_D^*, m_Y^*$. (iv) $\text{Cov}(h^*, m_D^*) \neq 0$.*

Part (i) is the collinearity limit. Part (ii) says the propensity retains curvature as it is pushed toward $\{0, 1\}$, and a nonlinear index driven toward a step function is the leading example, the one used in Section 6. Part (iv) is the genericity condition that makes the limit interesting. It holds whenever the nonlinear part of $m_D = \mathbb{E}[D \mid X]$ is not orthogonal to h^* . This is generic but not automatic. Writing $m_D(x) = q_0(x) + e(x)p(x)$ with $q_0(x) = \mathbb{E}[D(0) \mid X = x]$, the curvature ep inherited from the propensity can in principle be offset by curvature in the baseline q_0 , so (iv) rules out that knife-edge cancellation rather than following from relevance alone.

Proposition 2 (Collinearity contamination limit). *Under Assumptions 1–6, the signal terms vanish, $S_{D,t} \rightarrow 0$ and $S_{Y,t} \rightarrow 0$, while the contamination terms converge to $\text{Cov}(h^*, m_D^*) \neq 0$ and $\text{Cov}(h^*, m_Y^*)$. Hence*

$$\beta_{2SLS,t} \longrightarrow \frac{\text{Cov}(h^*, m_Y^*)}{\text{Cov}(h^*, m_D^*)}. \quad (5)$$

The limit is a ratio of covariate-level contrasts and carries no LATE interpretation. In particular it may lie outside $[\text{ess inf}_x \text{LATE}^(x), \text{ess sup}_x \text{LATE}^*(x)]$, and it may take the opposite sign of every $\text{LATE}^*(x)$.*

The reduced-form intuition is transparent. As $\text{Var}(Z \mid X) \rightarrow 0$, the within-covariate-cell experimental variation in the instrument disappears. All that survives in $\tilde{Z} = Z - \ell(X)$ is the projection error $h(X)$, which is a deterministic function of X . The estimator then

compares Y and D across covariate cells using weights $h(X)$, exactly the comparison an IV design is meant to avoid. The next result is the payoff for practice.

Corollary 1 (The conditional F does not warn). *The population Sanderson–Windmeijer conditional first-stage slope is $\pi_t = \mathbb{E}[\tilde{Z}_t D] / \text{Var}(\tilde{Z}_t) = (S_{D,t} + \theta_{D,t}) / \text{Var}(\tilde{Z}_t)$, and under Assumption 6,*

$$\pi_t \longrightarrow \frac{\text{Cov}(h^*, m_D^*)}{\mathbb{E}[(h^*)^2]} \neq 0.$$

Consequently the conditional first-stage F statistic, whose population counterpart is proportional to $n \pi_t^2 \text{Var}(\tilde{Z}_t) / \sigma_v^2$, is bounded away from zero and diverges with the sample size along the sequence, even though by Proposition 2 the estimand is pure contamination. A large conditional F therefore places no bound on the contamination bias (4).

Corollary 1 is the formal statement of the applied warning. The conditional F of Sanderson and Windmeijer (2016) correctly certifies that $\tilde{Z}$ is a strong predictor of D . Under collinearity, however, $\tilde{Z}$ predicts D through the contamination channel $h(X)$, not through the complier variation $e(1-e)p$. Strength and interpretability are not merely distinct. They can be driven by the same nuisance.

The following result shows the failure is not special to the conditional F but is shared by every strength diagnostic, and explains why. Such diagnostics are computed from the first stage, whereas the bias lives in the outcome.

Proposition 3 (Strength diagnostics cannot detect contamination). *Let T be any diagnostic that is a measurable functional of the joint distribution of (Z, D, X) , in particular any first-stage relevance or strength statistic, including the conditional F , the effective F of Montiel Olea and Pflueger (2013), and the Cragg–Donald statistic. Fix any such law with nonzero curvature, $h \neq 0$. Then T is uninformative about the contamination bias. There exist data-generating processes with that common joint distribution of (Z, D, X) , hence identical values of T , but arbitrarily different biases $\beta_{2SLS} - \bar{L}$. Detecting the bias requires the conditional distribution of Y . The restriction to $h \neq 0$ is not a weakening. When $h \equiv 0$ the first stage*

already certifies $\theta_D = \theta_Y = 0$ and hence no contamination, by Lemma 1, so the interesting case is exactly the one T cannot resolve.

Proposition 3 is the deep reason the applied bridge fails. No refinement of the first-stage report, however sophisticated, can be made to carry information about the contamination. By (4) the bias depends on $\theta_Y = \mathbb{E}[h(X)Y]$ and on $\bar{L} = S_Y/S_D$, both functionals of the outcome that are left completely free once the law of (Z, D, X) is fixed. This is an instance of the *level dependence* of Blandhol et al. (2022). Concretely, the free direction is the outcome level of the units the instrument does not move. Shifting the always- and never-taker outcome levels in the curvature direction h changes θ_Y , and hence the bias. It leaves the conditional LATEs and the entire law of (Z, D, X) unchanged, and with them every strength diagnostic. A strength statistic summarizes how the instrument moves compliers, whereas the bias lives with the non-compliers, which is why strength cannot see it. A valid diagnostic must therefore use Y . One thing the first stage *can* certify is the absence of curvature: by Lemma 1, $h \equiv 0$, a propensity linear in X and testable from (Z, X) alone by a RESET, forces $\theta_D = \theta_Y = 0$ and hence no contamination. The impossibility is therefore conditional on nonzero curvature. Once $h \neq 0$, no first-stage functional can say whether that curvature is harmful, because harm runs through θ_Y and $\bar{L}$, which only the outcome pins down. This is the precise content of the blanket statements elsewhere that a strong first stage cannot license a causal reading. This cuts in the opposite direction to the familiar impossibility that instrument *validity* cannot be confirmed even with infinite data (Słoczyński et al., 2026). There the outcome is of no help, while here it is exactly what a diagnostic requires. The observation both motivates and disciplines the test of the next section, whose central moment $\theta_Y = \mathbb{E}[h(X)Y]$ is precisely such an outcome functional. It also implies, a fortiori, that a Ramsey RESET applied to the first stage, a functional of (Z, X) alone, cannot distinguish a biased design from an unbiased one. We confirm this in Section 6.

4. A directed test for contamination

4.1. The null and its interpretation

By Lemma 1, $\beta_{2SLS} = \bar{L}$ whenever $\theta_D = \theta_Y = 0$. We therefore test

$$H_0 : \theta_D = 0 \text{ and } \theta_Y = 0 \quad \text{against} \quad H_1 : (\theta_D, \theta_Y) \neq (0, 0). \quad (6)$$

Under H_0 the linear-covariate 2SLS estimand retains its convex-weighted LATE interpretation. Because $\theta_D = \mathbb{E}[h(X)D]$ and $\theta_Y = \mathbb{E}[h(X)Y]$ depend only on the propensity curvature and on observables, H_0 is a statement about the *reduced form*. The consequential misspecification is nonlinearity of the instrument propensity $\mathbb{E}[Z \mid X]$ in the directions that also load on D and Y . This is what distinguishes the test from a Ramsey RESET applied to the first stage $D \sim (Z, X)$. RESET rejects for propensity curvature in *any* direction, including curvature orthogonal to the outcome that leaves (4) unaffected. This screening statistic instead probes only curvature that loads on D and Y , and does so without touching an endogenous regressor. It is still a *sufficient*-condition test. It can reject when the loadings satisfy $\theta_Y = \bar{L}\theta_D$ and the estimand is in fact unbiased (harmless curvature), which is why Section 4.3 adds the necessary-and-sufficient bias-targeting variant that probes the single direction $Y - \bar{L}D$.

4.2. Estimator and limiting distribution

Let $b(X) = (b_1(X), \dots, b_K(X))'$ be a vector of basis functions (for example polynomials or splines) orthogonalized against $(1, X')$ in the population, so that $\mathbb{E}[b(X)(1, X')] = 0$. These span the curvature directions to be probed. Model the propensity as $e(X) = \alpha_0 + X'\alpha_1 + b(X)'\delta + \rho(X)$ with $\mathbb{E}[\rho(X)b(X)] = 0$, so that the component of the curvature captured by the basis is $h_K(X) = b(X)'\delta$. Given an i.i.d. sample $\{(Y_i, D_i, Z_i, X_i)\}_{i=1}^n$, the test is computed in three steps.

- (i) Regress Z_i on $(1, X_i', b(X_i)')$ by OLS and form $\hat{h}_i = b(X_i)'\hat{\delta}$.

- (ii) Form the sample contamination moments $\hat{\theta}_D = n^{-1} \sum_i \hat{h}_i D_i$ and $\hat{\theta}_Y = n^{-1} \sum_i \hat{h}_i Y_i$, stacked as $\hat{\theta} = (\hat{\theta}_D, \hat{\theta}_Y)'$.
- (iii) Compute the Wald statistic $W_n = n \hat{\theta}' \hat{\Sigma}^{-1} \hat{\theta}$, where $\hat{\Sigma}$ is a consistent estimator of the asymptotic variance Σ of $\sqrt{n} \hat{\theta}$.

Assumption 7 (Test regularity). (i) $\{(Y_i, D_i, Z_i, X_i)\}_{i=1}^n$ are i.i.d. with $\mathbb{E}[Y^4] < \infty$, $\mathbb{E}[D^4] < \infty$ and $\mathbb{E}\|b(X)\|^4 < \infty$. (ii) $G = \mathbb{E}[b(X)b(X)']$ is nonsingular and the basis dimension K is fixed. (iii) $(\alpha_0, \alpha_1, \delta)$ are the unique population least-squares coefficients in $e(X) = \alpha_0 + X'\alpha_1 + b(X)'\delta + \rho(X)$ with $\mathbb{E}[\rho(X)(1, X', b(X)')'] = 0$. (iv) $\Sigma = \text{Var}(\psi_i)$ is nonsingular.

Proposition 4 (Limiting distribution). Under Assumption 7, $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Sigma)$, where Σ is the asymptotic variance of the two-step moment estimator with influence function

$$\psi_i = \begin{pmatrix} \hat{h}\text{-part for } D \\ \hat{h}\text{-part for } Y \end{pmatrix} = \begin{pmatrix} b(X_i)'\delta D_i - \theta_D \\ b(X_i)'\delta Y_i - \theta_Y \end{pmatrix} + \left(\mathbb{E}[D b(X)'], \mathbb{E}[Y b(X)'] \right)' G^{-1} b(X_i) u_i,$$

with $G = \mathbb{E}[b(X)b(X)']$ and $u_i = Z_i - \alpha_0 - X_i'\alpha_1 - b(X_i)'\delta$. Under H_0 , $W_n \xrightarrow{d} \chi_2^2$. Against a fixed alternative with $(\theta_D, \theta_Y) \neq 0$, $W_n \rightarrow \infty$, so the test is consistent. Against local alternatives $\theta = \eta/\sqrt{n}$ it has nontrivial power with noncentrality $\eta'\Sigma^{-1}\eta$.

The estimator $\hat{\theta}$ is a two-step GMM estimator, a first-step OLS for $(\alpha_0, \alpha_1, \delta)$ followed by the second-step sample moments (ii). Under Assumption 7, Newey and McFadden (1994, Theorem 6.1) give asymptotic linearity with the influence function above and $\Sigma = \text{Var}(\psi_i)$. The second term of ψ_i is the generated-regressor correction for first-step estimation of δ , and it is nonzero precisely because $b(X)$ enters through $\hat{\delta}$. The plug-in $\hat{\Sigma} = n^{-1} \sum_i \hat{\psi}_i \hat{\psi}_i'$, formed from the sample analogues $\hat{\psi}_i$, is consistent by the law of large numbers and the continuous mapping theorem, so $W_n \xrightarrow{d} \chi_2^2$ under H_0 . Because $\hat{\theta}$ is a Hadamard-differentiable functional of the empirical measure, the nonparametric pairs bootstrap is first-order valid by the functional delta method (van der Vaart, 1998, Theorem 23.9). Resampling (Y_i, D_i, Z_i, X_i)

jointly reproduces the generated-regressor term automatically, and we use it throughout Section 6.

The null (6) is a statement about the reduced form, and the estimator $\hat{\theta}$ never divides by the first-stage covariance $\mathbb{E}[\tilde{Z}D]$. The test is therefore immune to the weak-instrument problem, which is a small-denominator problem for β_{2SLS} . This is what separates the diagnostic from any procedure built on an IV estimate, and because the collinearity limit is precisely where it matters, we state it formally rather than leave it to Remark 3.

Proposition 5 (Size under weak identification, power under manufactured strength). *Call $\{P_n\}$ a weak-identification null sequence if: the fourth moments of $(Y, D, \|b(X)\|)$ are bounded uniformly in n and $\Sigma_n \rightarrow \Sigma_* \succ 0$ (so Assumption 7 holds uniformly along the sequence); the first-stage strength drifts to zero, $S_{D,n} \rightarrow 0$, as when $\text{Var}(Z | X) \rightarrow 0$ or $p(X) \rightarrow 0$; the curvature stays nondegenerate, $h_n \rightarrow h^* \neq 0$; and the null is maintained, $\theta_{D,n} = \theta_{Y,n} = 0$ for all n . Along any such sequence, under H_0 , $W_n \xrightarrow{d} \chi_2^2$, so the test that rejects when $W_n > \chi_{2,1-\alpha}^2$ has asymptotic size α regardless of the degree of identification; the same holds for the orthogonalized statistic W_n^o of Proposition 6. By contrast, the contaminated collinearity sequence of Assumption 6 has $\text{Cov}(h^*, m_D^*) \neq 0$, hence $\theta_{D,n} \not\rightarrow 0$, and is an alternative rather than a null. Along it the noncentrality $n\theta_n'\Sigma_n^{-1}\theta_n$ diverges and the test's power tends to one, even as every first-stage strength diagnostic diverges with the sequence (Corollary 1).*

The mechanism is that $\theta = (\theta_D, \theta_Y)$ and its influence function ψ are functionals of the reduced form (Z, D, Y, X) , estimated at rate $\sqrt{n}$ by the regression of Z on $(1, X', b(X)')$, whose behaviour is governed by $G = \mathbb{E}[bb']$ rather than by the relevance of the instrument. Because the curvature stays nondegenerate, $h_n \rightarrow h^* \neq 0$, the leading influence term $b(X)'\delta W - \theta_W$ stays nondegenerate and Σ_n remains nonsingular even as the generated-regressor correction shrinks with the residual variance of Z . The screening test thus retains its size in exactly the regime, $S_D \rightarrow 0$, in which every first-stage strength diagnostic diverges (Corollary 1) and a saturate-and-compare Hausman contrast collapses. The point is that weak identification and contamination are separate: the null sequence has $S_D \rightarrow 0$ without contamination, while

Assumption 6 couples $S_D \rightarrow 0$ with $\theta_D \not\rightarrow 0$ and is precisely the design the test is meant to reject. The bias-targeting refinement of Section 4.3 does require S_D bounded away from zero, for the separate reason that it plugs in $\hat{\bar{L}}$. That boundary is discussed there.

Proposition 4 fixes the basis dimension K , so the curvature is a parametric first step. Applied users will often prefer to estimate the propensity flexibly or by machine learning, which breaks Assumption 7(ii). The first-step error is then no longer $\sqrt{n}$ -negligible, and a naive plug-in is biased at rate faster than $\sqrt{n}^{-1}$. The following result restores valid inference by replacing the plug-in moment with a Neyman-orthogonal one, so that only $n^{1/4}$ -consistent, cross-fitted nuisances are required.

Proposition 6 (Orthogonalized test with a data-driven propensity). *Let the nuisances be the propensity $e(\cdot)$ and the level functions $m_W(\cdot) = \mathbb{E}[W \mid X]$, $W \in \{D, Y\}$, and write $\tilde{m}_W(X) = m_W(X) - L(m_W \mid 1, X)$ for the level function detrended of its affine part. For $W \in \{D, Y\}$ define the orthogonalized moment*

$$\psi_W^o(O; e, m_W) = h(X)W + \tilde{m}_W(X)(Z - e(X)) - \theta_W, \quad h = e - L(e \mid 1, X), \quad (7)$$

which has mean zero at the truth and is Neyman-orthogonal. Its pathwise derivative with respect to e and with respect to m_W vanishes. Estimate (e, m_D, m_Y) by any method with K -fold cross-fitting, and form $\hat{\theta}^o = (\hat{\theta}_D^o, \hat{\theta}_Y^o)'$, $\hat{\theta}_W^o = n^{-1} \sum_i \{\hat{h}(X_i)W_i + \hat{\tilde{m}}_W(X_i)(Z_i - \hat{e}(X_i))\}$, and $W_n^o = n \hat{\theta}^{o'} \hat{\Sigma}_o^{-1} \hat{\theta}^o$. Suppose the cross-fitted nuisances satisfy $\|\hat{e} - e\|_{L^2} = o_p(n^{-1/4})$ and $\|\hat{m}_W - m_W\|_{L^2} = o_p(n^{-1/4})$, hence the product rate $\|\hat{e} - e\|_{L^2} \|\hat{m}_W - m_W\|_{L^2} = o_p(n^{-1/2})$, with the fourth moments of Assumption 7(i) and $\Sigma_o = \text{Var}(\psi_i^o)$ nonsingular. Then $\sqrt{n}(\hat{\theta}^o - \theta) \xrightarrow{d} N(0, \Sigma_o)$ and, under H_0 , $W_n^o \xrightarrow{d} \chi_2^2$.

Remark 1 (Undersmoothed series as the intermediate case). The fixed-basis test of Proposition 4 and the orthogonalized test of Proposition 6 are two ends of a spectrum. An intermediate route keeps the plug-in moment of Section 4 but lets $K \rightarrow \infty$ under undersmoothing $K = o(n^{1/2})$ with the series two-step correction of Newey and McFadden (1994). It is orthog-

onalization (7) that removes the $n^{1/4}$ bottleneck and licenses off-the-shelf machine learning for the propensity. We use a fixed cubic basis throughout and treat K as a robustness dimension.

Remark 2 (What a non-rejection certifies). The fixed-basis screen of Proposition 4 has power only against contamination that projects onto the span of $b(X)$. Its noncentrality loads on $\mathbb{E}[b(X)D]$ and $\mathbb{E}[b(X)Y]$, so any curvature h orthogonal to b leaves the statistic unaffected. A non-rejection therefore certifies only that there is no detectable contamination *in the probed directions*, not that $\theta_D = \theta_Y = 0$ in every direction. This is the price of directing the test. The growing-basis and orthogonalized versions (Remark 1, Proposition 6) restore consistency against all directions as $K \rightarrow \infty$. With the fixed cubic basis used below, the empirical “clears” of Section 7 should accordingly be read as “no contamination detectable at cubic order,” and K as a robustness dimension one can raise.

Remark 3 (Why not just saturate and compare?). The natural alternative to our test is to estimate β under a saturated covariate specification and compare it to the linear-specification estimate via a Hausman contrast. Under collinearity this is exactly the wrong tool. Saturating drives the effective instrument toward the vanishing signal term $S_D \rightarrow 0$ (Proposition 2), so the saturated estimator is weakly identified and its sampling variability swamps the contrast. Our test sidesteps this because it never forms an IV ratio, and its size is controlled along the collinearity sequence (Proposition 5), the property the saturate-and-compare contrast cannot have.

4.3. A bias-targeting variant

The null (6) is *sufficient* for $\beta_{2SLS} = \bar{L}$ but not necessary. By (4) the bias vanishes whenever $\theta_Y = \bar{L}\theta_D$, which can hold with $(\theta_D, \theta_Y) \neq 0$. The leading example is *harmless curvature*. Here the propensity is nonlinear in X , so $h \neq 0$ and, since m_D inherits the curvature, $\theta_D \neq 0$, yet the outcome level is linear in X and the effect is homogeneous, so $\theta_Y = \bar{L}\theta_D$ and the estimand is unbiased. Testing (6) then rejects a design that is in fact fine.

The screening test of Section 4 is therefore best read as a strongly identified, conservative first pass.

The exact no-bias restriction is a single moment. Since $\theta_Y - \bar{L}\theta_D = \text{Cov}(h(X), m_Y(X) - \bar{L}m_D(X)) = \mathbb{E}[h(X)\{Y - \bar{L}D\}]$,

$$H_0^* : \mathbb{E}[h(X)\{Y - \bar{L}D\}] = 0, \quad (8)$$

which is necessary and sufficient for $\beta_{2SL} = \bar{L}$ under Assumption 3. Written as a ratio, H_0^* invites a plug-in of $\bar{L} = S_Y/S_D$, fragile precisely where contamination is worst. Clearing the denominator gives an equivalent restriction in objects that are all $\sqrt{n}$ -estimable irrespective of the signal. With $\psi \equiv \theta_Y S_D - \theta_D S_Y$,

$$b = \beta_{2SL} - \bar{L} = \frac{\psi}{S_D \mathbb{E}[\tilde{Z}D]}, \quad H_0^* \iff \psi = 0. \quad (9)$$

The two forms test the same null. They differ in where the small signal S_D appears, and that difference is a genuine trade-off rather than a matter of taste. A statistic that tracks b directly (the plug-in T_n^* below) has full power but, dividing implicitly by $\hat{S}_D$, loses pivotality as $S_D \rightarrow 0$. The ψ -form is pivotal even at that boundary but, since $\psi = b S_D \mathbb{E}[\tilde{Z}D]$ vanishes with the signal, it trades power for size there. Neither escapes the trade-off, because (9) forces the choice. This is why the recommendation below to report both tests is a *consequence* of the identity, not a hedge. The screening test of Section 4 keeps its size in exactly the regime where the bias-targeting test loses pivotality, and the bias-targeting test keeps power in exactly the regime of harmless curvature, where the screen false-alarms. In the finite-sample designs of Section 6.4 the pairs bootstrap, which recomputes $\hat{\bar{L}}$ on each resample and thereby effectively clears the denominator, delivers the pivotal behaviour while retaining power across the collinearity range (Table B.8). We estimate H_0^* by plugging in the bias-corrected $\hat{\bar{L}}$ of (11) (Section 5) and the flexible curvature $\hat{h}_i$. This forms $\hat{g} = n^{-1} \sum_i \hat{h}_i \{Y_i - \hat{\bar{L}}D_i\}$ and the statistic $T_n^* = \hat{g}^2/\hat{v}$, where $\hat{v}$ is the pairs-bootstrap variance of $\hat{g}$ itself, obtained by

recomputing $\hat{h}$ and $\hat{\bar{L}}$ on each resample. Suppose H_0^* and Assumption 7 hold, and S_D is bounded away from zero in the fixed data-generating process, so that $\hat{\bar{L}}$ is $\sqrt{n}$ -consistent for $\bar{L}$ (Proposition 7). Then the delta method gives $\sqrt{n}\hat{g} \xrightarrow{d} N(0, v)$ for a finite $v > 0$, and the pairs-bootstrap variance tracks the sampling variance of $\hat{g}$, so that $n\hat{v} \xrightarrow{p} v$. Hence $T_n^* = \hat{g}^2/\hat{v} = n\hat{g}^2/(n\hat{v}) \xrightarrow{d} \chi_1^2$, and the test is consistent against any fixed alternative with nonzero bias. The requirement that S_D be bounded from zero is exactly where the collinearity limit bites. As $S_D \rightarrow 0$ the plug-in $\hat{\bar{L}}$ ceases to be $\sqrt{n}$ -consistent and T_n^* is no longer asymptotically pivotal (Proposition 8). This is the size side of the trade-off in (9). We do not prove that the pairs bootstrap restores pivotality at the $S_D \rightarrow 0$ boundary. In the finite-sample designs of Table B.8 it holds size across the collinearity range, but that is simulation evidence over a bounded grid, not a weak-signal validity result, so the bias-targeting test carries a demonstrated guarantee only in the strongly identified regime. The protocol is therefore to report both tests, the strongly identified screen of Section 4 and the bias-targeting confirmatory test (8), and to read them alongside the contamination share $\hat{\varrho}$, which by Corollary 2 says how much identification a correction would cost. Section 6.4 compares both against a RESET.

4.4. Reporting the contamination share

Beyond a hypothesis test, the decomposition yields a single interpretable number. Define the *contamination share*

$$\varrho \equiv \frac{\theta_D}{S_D + \theta_D} = \frac{\theta_D}{\mathbb{E}[\tilde{Z}D]}, \quad (10)$$

the fraction of the observed first-stage *covariance* $\mathbb{E}[\tilde{Z}D]$ attributable to propensity curvature rather than to complier variation. It is a share of the covariance, not literally of the conditional F or the concentration parameter, which are quadratic in that covariance and also involve the residualized-instrument variance. We read ϱ as a decomposition of the first-stage covariance and, through Corollary 2, of the identifying variation a correction surrenders, and we use “apparent strength” only in that loose sense. In the nonlinear-propensity family

of Section 6, (10) reduces to $\varrho = \mathbb{E}[h^2]/\text{Var}(\tilde{Z})$, the share of the residualized-instrument variance that is curvature, and there $\theta_D > 0$ so $\varrho \in [0, 1]$. In general ϱ need not be a proper fraction. Since $S_D > 0$ always but θ_D is a signed covariance, $\varrho < 0$ when the curvature deflates the first-stage covariance ($\theta_D < 0$), $\varrho > 1$ when it reverses its sign ($\theta_D < -S_D$), and ϱ is unstable when signal and curvature nearly cancel ($\mathbb{E}[\tilde{Z}D] \approx 0$). The share reading, and the statement that curvature *inflates* the conditional F , hold under the sign condition $\theta_D > 0$ that the mode-(a) designs here satisfy; outside it $\theta_D < 0$ instead deflates the F , and ϱ is best read as a signed decomposition of the first-stage covariance and reported with its sign. It is estimated by $\hat{\varrho} = \hat{\theta}_D / \widehat{\mathbb{E}[\tilde{Z}D]}$ with $\widehat{\mathbb{E}[\tilde{Z}D]} = n^{-1} \sum_i \tilde{Z}_i D_i$, and a percentile bootstrap furnishes a confidence interval. We recommend reporting $\hat{\varrho}$ next to the conditional F . A large F paired with a non-negligible $\hat{\varrho}$ signals that the strength is partly manufactured, the signature of mode (a). Crucially, ϱ is a first-stage gauge and measures only mode (a). It is *not* a bias estimate, because by Proposition 3 the bias also loads on the outcome-side term θ_Y . A small $\hat{\varrho}$ therefore does not certify a design. When it is paired with a directed-test rejection, that combination is itself the signature of mode (b), namely honest strength with outcome-side bias, as in the 401(k) design, where $\hat{\varrho} = 0.03$ yet the estimand is off by a third. The share thus complements, and does not replace, the outcome-based tests of Sections 4–4.3. Read together, a large F , the share $\hat{\varrho}$, and the directed-test p -value locate which mode, if either, is in play.

5. Bias-corrected estimation and the identification trade-off

The decomposition suggests an immediate correction, namely to subtract the estimated contamination from numerator and denominator. Using $\mathbb{E}[\tilde{Z}Y] - \theta_Y = \mathbb{E}[(Z - e(X))Y]$ and likewise for D , the corrected estimand equals the signal ratio $\bar{L}$. It is implemented by replacing the linear-projection residual $\tilde{Z}$ with the residual from a *flexibly* estimated propensity $\hat{e}(\cdot)$,

$$\hat{\bar{L}} = \frac{\sum_i (Z_i - \hat{e}(X_i)) Y_i}{\sum_i (Z_i - \hat{e}(X_i)) D_i}. \quad (11)$$

Proposition 7 (Consistency of the corrected estimator). *Suppose $\hat{e}(\cdot)$ satisfies $\|\hat{e} - e\|_{L^2} = o_p(1)$ and, in the fixed data-generating process, $S_D = \mathbb{E}[e(X)\{1 - e(X)\}p(X)] \neq 0$. Then under Assumptions 1–5, $\hat{\bar{L}} \xrightarrow{p} \bar{L} = S_Y/S_D$, the convex-weighted average of conditional LATEs with weights proportional to $e(x)\{1 - e(x)\}p(x)$. Consistency requires no orthogonality or rate condition. Asymptotic normality and inference do (see the Appendix and Remark 1), and, since S_D may be small, weak-instrument-robust confidence sets are appropriate.*

Estimator (11) is the residualized-propensity, or partialling-out, IV estimator. It coincides with the recommendation of Blandhol et al. (2022) to saturate the covariates, here written in influence-function form. Our contribution is not the estimator but the following statement of its cost, which the collinearity limit makes precise.

Proposition 8 (No free lunch). *The corrected estimator’s own first-stage covariance is exactly the signal term, $\mathbb{E}[(Z - e(X))D] = S_D = \mathbb{E}[e(X)\{1 - e(X)\}p(X)]$. Under the collinearity sequence of Assumption 6, $S_D \rightarrow 0$. Hence the corrected estimator is weakly identified in precisely the regime in which the correction matters most. There is no specification of the covariates that both removes the contamination and preserves the apparent first-stage strength, because the strength was the contamination.*

Corollary 2 (The identification surrendered equals the contamination removed). *The corrected and reported (contaminated) first-stage covariances obey the exact identity*

$$\underbrace{S_D}_{\text{corrected first stage}} = (1 - \varrho) \underbrace{\mathbb{E}[\tilde{Z}D]}_{\text{reported first stage}}, \quad \varrho = \frac{\theta_D}{\mathbb{E}[\tilde{Z}D]},$$

where ϱ is the contamination share (10). De-contamination scales the identifying covariance down by exactly ϱ , and the corrected concentration parameter $S_D^2/\{\sigma_v^2 \mathbb{E}[e(1 - e)]\}$ carries a factor $(1 - \varrho)^2$ in its numerator. The trade-off of Proposition 8 is therefore quantitative and known in advance. The fraction of identifying variation surrendered by correcting is the same ϱ that measured how much of the apparent strength was contamination. A design with

$\varrho \approx 0$, such as the 401(k) instrument of Section 7.1 ($\hat{\varrho} = 0.03$), is corrected at negligible cost, whereas a design whose strength is mostly curvature ($\varrho \rightarrow 1$) cannot be both corrected and strongly identified.

The practical message follows. The analyst facing a strong conditional F under instrument-covariate collinearity has not found a strong design that merely needs a better second-stage specification. The strength is an artifact of the very nonlinearity that biases the estimand, and removing it returns the design to (correctly) weak. The response is to diagnose with our test, followed by either a redesigned instrument or explicit weak-IV-robust inference on the corrected estimand (Andrews et al., 2019; Montiel Olea and Pflueger, 2013).

6. Monte Carlo evidence

6.1. Design

Throughout, $X \sim \text{Uniform}(-2, 2)$ and $g(X) = X^2 - \frac{4}{3}$ (mean zero). Potential treatments satisfy monotonicity via a latent uniform V , with $D(1) = \mathbf{1}\{V \leq \pi_0 + \Delta\}$ and $D(0) = \mathbf{1}\{V \leq \pi_0\}$, where $\pi_0 = 0.2$ and the conditional first stage is $\Delta = p(X) = 0.4$. The conditional LATE is heterogeneous and strictly positive, $\text{LATE}(X) = 1 + 0.25X \in [0.5, 1.5]$, and the outcome is $Y = A_\mu g(X) + \text{LATE}(X)D + \mathcal{N}(0, 1)$. Two instrument-propensity families are used. The first is a *nonlinear* logit propensity $e(X) = \Lambda(\kappa g(X))$, whose curvature knob κ indexes collinearity, with larger κ driving $\text{Var}(Z \mid X) \rightarrow 0$. The second is a *linear-probability* propensity $e(X) = 0.5 + aX + b g(X)$ with $a = 0.08$, clipped to $[0.02, 0.98]$, whose curvature knob b switches contamination *off* at $b = 0$, where the propensity is affine and the null holds, and *on* for $b \neq 0$. So $b = 0$ is exactly the no-contamination null and $|b|$ indexes the induced bias. This presumes the clip is slack. It is at the moderate a, b used here, so no curvature is manufactured by clipping at $b = 0$. All numbers below are from the **R** scripts released with the paper.

Table 1: Collinearity limit. 2SLS estimand vs. the conditional F ($n = 200\,000$, $A_\mu = 1.5$). True $\text{LATE}(X) \in [0.5, 1.5]$ throughout.

κ	β_{2SLS}	cond. F	contam. limit	$\mathbb{E}[e(1 - e)]$
0.10	1.527	39 808	151.4	0.249
0.25	2.316	39 888	61.9	0.245
0.50	3.497	40 459	32.8	0.230
1.00	5.408	38 776	19.1	0.189
2.00	7.325	38 904	13.1	0.118
4.00	8.487	37 790	10.7	0.057
8.00	8.794	37 881	9.77	0.027
16.0	8.877	37 979	9.32	0.013
32.0	8.860	38 215	9.11	0.007

6.2. The collinearity limit

Table 1 traces the nonlinear-propensity design ($A_\mu = 1.5$) as κ rises, with $n = 200\,000$ to approximate the population. The 2SLS estimand leaves the LATE hull $[0.5, 1.5]$ already at mild collinearity and converges monotonically to the theoretical contamination limit $\text{Cov}(h, m_Y) / \text{Cov}(h, m_D)$ of Proposition 2. The last-column conditional variance $\mathbb{E}[e(1 - e)]$ collapses toward zero as predicted. Throughout, the conditional first-stage F remains on the order of 40 000, four orders of magnitude above the conventional threshold of 10, confirming Corollary 1. Figure 1 plots the two panels.

6.3. Size and power

Two exercises confirm the test’s basic behaviour. Both are collected in Appendix B. On a single large contaminated dataset ($n = 100\,000$, $\kappa = 4$) the directed test rejects at $p < 10^{-4}$ while the conditional F is 20 723, and the bias-corrected estimator recovers the true convex-weighted LATE (Table B.6). Over 300 replications at $n = 3000$ it has a conservative size of 1.3% against the 5% level under a linear propensity, and power essentially one under a curved one, at a mean conditional F near 575 in *both* cases. The F cannot discriminate the two designs the test separates almost perfectly. A power sweep (Table B.7, Figure B.3) raises the induced bias from 0 to 2.68 while the mean conditional F stays pinned near 1740.

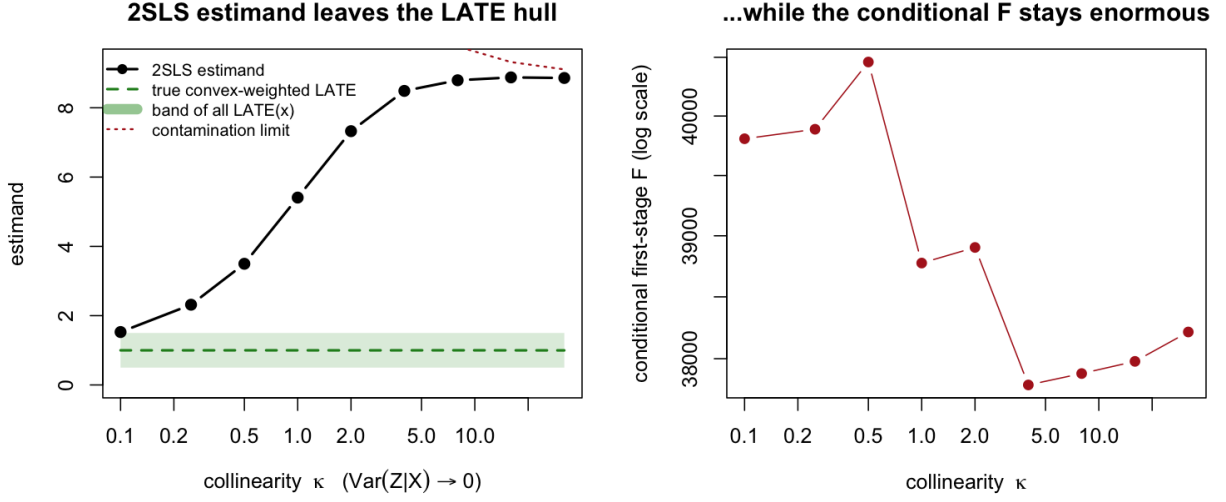


Figure 1: Left panel. As collinearity κ rises, the 2SLS estimand (solid) leaves the band of all conditional LATEs (shaded) and departs from the true convex-weighted LATE (dashed), converging to the contamination limit (dotted). Right panel. The conditional first-stage F (log scale) stays near 40 000 throughout, far above the weak-IV threshold of 10.

6.4. Head-to-head comparison across three designs

Table 2 and Figure 2 compare three procedures across three designs at $n = 3000$. The first procedure is a Ramsey RESET testing curvature of the propensity $Z \sim X$, a functional of (Z, X) alone. The second is the strongly identified screening test of Section 4 ($H_0 : \theta_D = \theta_Y = 0$), which we label “Directed-S”. The third is the bias-targeting test (8) of Section 4.3, labelled “Directed-X”. The *contaminated* design has a nonlinear propensity and a nonlinear outcome, so it carries real bias. The *harmless* design has a nonlinear propensity but a linear outcome and a homogeneous effect, so the bias is zero despite strong curvature. The *clean* design has a linear propensity and zero bias. All three designs carry a mean conditional F near 600.

The pattern illustrates Proposition 3. In the harmless design the RESET rejects 100% of the time. It detects the propensity curvature but, being blind to the outcome, cannot tell that the curvature is inconsequential, and the same $F \approx 600$ that accompanies a large bias in the contaminated design accompanies zero bias here. Directed-S also false-alarms (100%), because it tests the sufficient, not the necessary, condition. Only Directed-X separates the three designs as the theory requires. It has full power against genuine contamination, yet

Table 2: Head-to-head rejection rates (5% level, 300 replications, $n = 3000$, cubic basis, $B = 99$). “mean cond. F ” is the average Sanderson–Windmeijer conditional first-stage F .

design	RESET	Directed-S	Directed-X	mean cond. F
CONTAMINATED (bias $\neq 0$)	1.00	1.00	1.00	602
HARMLESS (bias = 0)	1.00	1.00	0.047	602
CLEAN (bias = 0)	0.06	0.013	0.083	572



Figure 2: Rejection rates of the RESET, the sufficient (Directed-S) and the bias-targeting (Directed-X) tests across the contaminated, harmless, and clean designs. In the harmless design (middle group) the RESET and Directed-S false-alarm while Directed-X holds size, and the conditional F (near 600 in all groups) cannot discriminate.

holds size (4.7%) in the harmless design that the other two procedures, and the conditional F , cannot certify. In the clean design its rejection rate is 8.3%, about 2.6 null standard errors above the 5% level at 300 replications, a mild finite-sample over-rejection that the larger- n power curve (Table B.7, where the $b = 0$ size falls to 0.5–1.0% by $n \geq 3000$) does not carry. We therefore read Directed-X as correctly sized against harmless curvature and mildly liberal on a small-sample clean design, not as exactly calibrated in finite samples.

Table B.8 in Appendix B pushes the same contrast along the collinearity knob. Both directed tests keep full power against genuine bias as $\text{Var}(Z \mid X)$ falls by an order of magnitude, so neither loses power to collinearity. They part company only in the harmless design. There Directed-S false-alarms every time, because its *sufficient* condition $\theta = 0$ is false whenever the propensity is nonlinear, while Directed-X controls size throughout, if anything

conservatively, the pairs bootstrap insulating it from the loss of pivotality that Corollary 2 anticipates. This is the content of the “report both” recommendation. Directed-S is a cheap screen that, at the cubic basis used here, misses no contamination in the probed directions but cannot vindicate harmless curvature, and Directed-X is the adjudicator that, in those same directions, separates harmful from harmless. Both guarantees are relative to the fixed basis, per Remark 2. A growing or orthogonalized basis extends them to all directions.

6.5. *Estimating the contamination share*

Table B.9 and Figure B.4 in Appendix B validate the share estimator (10). The plug-in $\hat{\varrho}$ at $n = 3000$ tracks the population share from $\varrho \approx 0.02$ to $\varrho \approx 0.75$ with 92–96% bootstrap coverage. Because the bias also loads on the outcome-side term, a modest share can coincide with substantial bias. At $\varrho = 0.02$ the estimand is already displaced by 1.3 here. Thus $\hat{\varrho}$ flags manufactured strength, while the test of Section 4.3 adjudicates its consequences.

7. Empirical application

We apply the diagnostic to a panel of canonical just-identified IV designs and report all of them. The panel spans the three behaviours the theory predicts. The first is a strong-instrument design whose linear-covariate estimate is materially biased and which the correction repairs (Section 7.1). The second is a set of designs with genuine but harmless propensity curvature that the screening test detects and the bias-targeting test correctly clears (Section 7.2). The third is a set of designs the diagnostic clears outright (Section 7.3). Throughout, the flexible propensity uses a low-order polynomial (or spline) basis and inference uses the pairs bootstrap of Section 4.

A scope caveat governs how these results should be read. The binary-instrument, binary-treatment model of Section 2 holds exactly for the headline 401(k) design (eligibility and participation are both binary) and for the Catholic and Card designs, which have binary instruments. The remaining designs use the instrument and treatment in their native multivalued or continuous form. Maimonides predicted and actual class size, the Cornwell–Trumbull

offense mix and arrest probability, and the schooling designs’ years of education are not dichotomized. The reduced-form objects the diagnostic actually computes, namely the curvature $h = e - L(e \mid 1, X)$, the contamination moments $\theta_D = \mathbb{E}[hD]$ and $\theta_Y = \mathbb{E}[hY]$, the share ϱ , and the two directed tests, are functionals of $\mathbb{E}[Z \mid X]$ and the level functions alone. They are therefore well defined and, by the arguments of Sections 4–5, retain their size and consistency for a general Z and D , since none of those arguments uses the binary structure. What the binary model buys is the *interpretation* of the signal ratio $\bar{L} = S_Y/S_D$ as a convex average of conditional LATEs, through the Wald–LATE identity of Lemma 1. For a multivalued instrument or treatment $\bar{L}$ is instead the saturated-propensity linear-IV estimand, a Blandhol-type weighted average of causal responses whose convexity requires the multivalued monotonicity conditions we do not impose here. For those designs the tests should accordingly be read as detecting, or certifying the absence of, functional-form contamination of the linear-covariate 2SLS estimand relative to its saturated-propensity counterpart, not as a statement about a convex-LATE object. The multivalued convex-LATE decomposition is the open frontier of Section 9.

7.1. A misleading strong first stage in the 401(k) design

Our headline application is the effect of 401(k) participation on household net financial assets, using 401(k) *eligibility* as an instrument for participation (Poterba et al., 1995; Abadie, 2003). We use the cross-section of $n = 9,275$ households of Wooldridge (2010). Eligibility is an extremely strong instrument, with a conditional first-stage F exceeding 7,000. But firms offer eligibility in a manner strongly nonlinear in income, and income is a first-order, nonlinear determinant of wealth. This is precisely the configuration of Lemma 1 in which linear covariate adjustment contaminates the estimand.

Table 3 reports the just-identified 2SLS estimate (participation on net financial assets, in thousands of dollars) under three income specifications, with the diagnostic.³ Under a

³Full specification, from the released `iv_401k.R`. Treatment D is 401(k) participation, instrument Z is 401(k) eligibility, outcome Y is net financial assets. The linear control vector is (income, age, married,

Table 3: The 401(k) design ($n = 9,275$). Effect of participation on net financial assets (\$000s), instrumented by eligibility. A first-stage $F > 7,000$ does not prevent contamination under linear income control, and the correction recovers the flexible-income answer. Test p -values from the pairs bootstrap. For the corrected row, cond. F is the corrected estimator’s own first stage (regressing D on the residualized instrument $Z - \hat{e}(X)$), the bracketed interval is the pairs bootstrap, and its identification-robust Anderson–Rubin 95% interval is [8.2, 17.0].

income control	2SLS effect	cond. F	$\hat{\varrho}$	Directed-S p	Directed-X p
linear (naive)	8.40	7401	0.03	< 0.0001	0.0001
quadratic	13.66	7423	0.00	0.121	0.113
natural splines (benchmark)	13.05	n/a	n/a	n/a	n/a
<i>contamination-corrected</i>	<i>12.6 [9.0, 16.0]</i>	<i>7295</i>			

linear income control, a common and parsimonious default, the estimate is 8.40. Both the screening and the bias-targeting tests reject decisively ($p < 10^{-3}$) *despite* $F > 7,000$, and the bias-corrected estimand is 12.6 ([9.0, 16.0]). The correction is validated in three ways. Controlling income *quadratically* removes the flag ($p_X = 0.11$) and gives 13.7. A flexible natural-spline income control gives 13.1. Our correction from the linear specification, 12.6, coincides with both. The naive linear-income estimate is the outlier, understating the effect by roughly a third. The rejection is robust to the curvature basis, since polynomial and natural-spline bases both give $p_X < 10^{-3}$, and it is self-validating, since it disappears exactly when income is controlled flexibly.

This is the paper’s thesis on real data. A first-stage F above seven thousand offers no protection. Under linear income control the estimand is contaminated by the covariance between the eligibility propensity’s income curvature and the steep income profile of wealth. By Proposition 3 no first-stage diagnostic could reveal this, and none does. Both numbers computed from the first stage, the conditional $F > 7,000$ and the contamination share $\hat{\varrho} = 0.03$, report a clean, strongly identified design. They miss the bias because here it

male, family size). The curvature basis $b(X)$ for the linear specification is $(\text{inc}^2, \text{inc}^3, \text{age}^2, \text{inc} \times \text{age})$ residualized against the linear controls. The quadratic specification adds $\text{inc}^2, \text{age}^2$ to the controls with basis $(\text{inc}^3, \text{inc}^4, \text{age}^3, \text{inc} \times \text{age})$. The spline robustness check uses a natural-spline income basis with six degrees of freedom, and the flexible benchmark controls income and age by natural splines with five and four degrees of freedom. Inference is a pairs bootstrap with $B = 1499$ resamples that refits the propensity, the curvature basis, and the level functions on each resample.

loads on the outcome side, through θ_Y rather than the first-stage term θ_D , the channel no relevance diagnostic can reach. Only the outcome-based directed test detects it, and the correction returns the specification-robust answer. The 401(k) design is thus failure mode (b) in near-pure form. The strength is honest and overwhelming, and with $\hat{\varrho} = 0.03$ the bias is *predominantly* on the outcome side. Decomposing the displacement of 8.4 from the corrected 12.6, the first-stage channel $-\bar{L} \hat{\varrho}$ accounts for only about a tenth of it and the outcome term θ_Y for the rest, so the first-stage numbers see essentially none of the bias. It is the empirical face of Proposition 3. The very design in which every first-stage number is reassuring is the one that is biased.

The corrected estimand still needs an interval. Proposition 8 warns that de-contamination can destroy identification, so a corrected point estimate must be read with an interval that is robust to weak identification. Here that caveat does not bind. Because the contamination share is only $\hat{\varrho} = 0.03$, the corrected estimator retains almost all of the first-stage covariance, and its *own* first-stage F (regressing D on the residualized instrument $Z - \hat{e}(X)$) is 7295. The corrected estimator is therefore strongly identified, and the identification-robust Anderson–Rubin 95% interval, $[8.2, 17.0]$, is bounded and coincides with the normal-theory interval, confirming that the pairs-bootstrap interval $[9.0, 16.0]$ is valid in this design. The general rule follows recommendation (3). When $\hat{\varrho}$ is small the corrected first stage survives and standard or bootstrap intervals apply, whereas when $\hat{\varrho}$ approaches one the corrected first stage collapses and the interval must be read off the Anderson–Rubin set rather than the bootstrap. The careful applied literature avoids the problem by controlling income flexibly, which is exactly what the correction reproduces. The contribution is a diagnostic that flags the common parsimonious specification that does not do so, and that a large F would otherwise excuse.

7.2. *Genuine but harmless curvature*

Two further strong-instrument designs show the value of separating the screening test from the bias-targeting test (Table 4). In the class-size design of Angrist and Lavy (1999), the

Table 4: Strong-instrument designs with genuine but harmless curvature. The screening test rejects (curvature present) but the bias-targeting test clears and the correction $\approx$ the naive estimate.

design	n	cond. F	$\hat{\rho}$	Directed-S p	Directed-X p	naive $\rightarrow$ corr.
Maimonides, math	2013	458	0.23	< 0.0001	0.19	$-0.171 \rightarrow -0.217$
Maimonides, verbal	2013	458	0.23	< 0.0001	0.75	$-0.221 \rightarrow -0.230$
Schooling, father’s educ.	741	95	0.15	0.001	0.18	$0.110 \rightarrow 0.119$

instrument is Maimonides’ rule predicted class size, a deterministic sawtooth in enrollment. The conditional F is 458 and the screening test rejects decisively ($\hat{\rho} = 0.23$, $p < 10^{-4}$), so a large part of the apparent strength is propensity curvature, invisible to the F . Yet the bias-targeting test does *not* reject ($p_X = 0.19$). The correction moves the math estimate from -0.171 to -0.217 . That shift is sizeable in proportional terms, but the bias-targeting p -value shows it is not statistically distinguishable from zero at the cubic basis, so the design is certified as harmless-curvature rather than as numerically unchanged. A schooling IV using father’s education (Wooldridge, 2010) behaves the same way, with a much smaller point shift. These are real-data instances of the harmless-curvature case of Section 6.4, read at cubic order per Remark 2.

7.3. Designs the diagnostic clears

Table 5 reports the remaining eight designs. Among them are schooling instrumented by college proximity (Card, 1995) and by mother’s education (Mroz, 1987), the institutions–income IV (Acemoglu et al., 2001) (inconclusive at $n = 64$), openness–inflation (Romer, 1993), and the Cornwell–Trumbull crime IV (Cornwell and Trumbull, 1994). All clear both tests, with a corrected estimand indistinguishable from the linear-covariate 2SLS estimate. The diagnostic does not manufacture false alarms. These non-rejections are at cubic order, and by Remark 2 certify only that no contamination is detectable in the probed directions, not its absence in every direction.

Across the panel the diagnostic behaves as designed. It flags and repairs a consequential contamination in a design with an overwhelming first stage (401(k), mode b), detects genuine curvature that it then correctly certifies harmless (Maimonides, schooling by father’s

Table 5: Designs the diagnostic clears (both tests $p > 0.10$, corrected $\approx$ naive). The contamination share $\hat{\varrho}$ is small except at AJR, which is reported as inconclusive given $n = 64$.

design	n	cond. F	$\hat{\varrho}$	Directed-S p	Directed-X p
Schooling, college proximity (Card)	3010	14	0.03	0.74	0.81
Schooling, mother’s educ. (Mroz)	428	73	0.02	0.83	0.74
Cigarette demand, taxes	48	326	0.02	0.85	0.79
Institutions, settler mortality (AJR)	64	12	0.21	0.19	0.98
Openness–inflation, log land (Romer)	114	44	0.00	0.97	0.79
Botswana fertility, birth timing	4358	43	0.00	0.91	0.97
Catholic schooling, parent Catholic	7430	389	0.01	0.12	0.25
Crime, offense mix (Cornwell–Trumbull)	630	105	0.00	0.56	0.31

education), and clears the remaining designs. That well-identified published designs largely clear the bias-targeting test is not evidence that the problem is rare. It is evidence that the diagnostic is *specific*, in that it does not manufacture alarms on clean designs, the property a RESET lacks (Section 6.4). The 401(k) case shows the converse, that a large first-stage F is no guarantee, and that the contamination it conceals is both detectable and correctable.

8. Recommendations for practice

Our results imply a short protocol that adds little cost to a standard IV analysis. (1) Report the *conditional* first-stage F of Sanderson and Windmeijer (2016), but do not treat it as evidence for a causal reading of the second stage, and report the estimated contamination share $\hat{\varrho}$ (10) beside it. (2) Estimate the instrument propensity $\mathbb{E}[Z \mid X]$ flexibly and test its curvature against D and Y using the screening statistic of Section 4 and, to distinguish harmful from harmless curvature, the bias-targeting test of Section 4.3. A screening rejection signals curvature that loads on D and Y even when the conditional F is large, but it does not by itself establish bias. Use the bias-targeting test to decide. (3) If the *bias-targeting* test rejects, report the bias-corrected estimand (11) together with weak-IV-robust inference, and recognize (Proposition 8) that a collapse in the corrected first stage means the apparent strength was contamination rather than identification. A screening rejection that the bias-targeting test does not confirm is the harmless-curvature case and calls for no correction. (4)

Where feasible, prefer an instrument whose propensity is approximately linear in the controls, or saturate the controls at the design stage while monitoring the resulting conditional strength.

9. Conclusion

The applied ritual of reporting a first-stage F answers whether an instrument moves the treatment, not whether the second-stage estimate is a meaningful causal average. Under instrument–covariate collinearity these questions come apart in a specific, detectable way. The same propensity curvature that contaminates the linear-covariate 2SLS estimand also inflates the conditional F , so the strength diagnostic cannot see the problem and the natural saturation fix trades the bias for weak identification. A directed conditional-moment test, built entirely from reduced-form regressions, has power exactly where the conditional F is silent.

The analysis is deliberately confined to a binary instrument, a binary treatment, and just identification. We now say what survives beyond that boundary and what does not. Overidentification is the benign direction, though it is less mechanical than a single reduced form. With a vector of instruments the FWL numerator and denominator of Lemma 1 each become a curvature–level covariance vector, one entry per instrument, but the 2SLS estimand combines them through the residualized-instrument second-moment matrix, so the population coefficient is a weighted combination of the per-instrument signal and contamination vectors rather than a clean term-by-term ratio. What does carry over exactly is the diagnostic content. The impossibility result (Proposition 3) holds verbatim, since the bias still loads on the outcome, and the directed test extends by stacking the moments $\mathbb{E}[h_j(X)D]$ and $\mathbb{E}[h_j(X)Y]$ across instruments j and testing them jointly. The stacked moment restriction is a sufficient screen for no contamination. An exact overidentified analogue of the bias-targeting restriction, which must invert the second-moment matrix, we leave to future work. Multiple or multivalued treatments are the substantive frontier. There our

single-treatment, functional-form contamination compounds with the *cross-treatment* contamination of Goldsmith-Pinkham et al. (2024), a conceptually distinct channel that persists even under a saturated propensity, and disentangling the two is genuinely open. What extends cleanly in every case is the reason the first stage cannot help. The bias always loads on the conditional law of the outcome, so Proposition 3 is not special to the binary model. The remaining loose end, a data-driven curvature basis, is closed by Proposition 6. We leave the multivalued extension to future work, where the object to be tested is a matrix rather than a scalar contamination.

Appendix A. Proofs

Proof of Lemma 1

By (1), $\beta_{2SLS} = \mathbb{E}[\tilde{Z}Y]/\mathbb{E}[\tilde{Z}D]$ with $\tilde{Z} = Z - \ell(X)$. Write $\tilde{Z} = (Z - e(X)) + h(X)$ using $h = e - \ell$. For any integrable $W \in \{D, Y\}$,

$$\mathbb{E}[\tilde{Z}W] = \mathbb{E}[(Z - e(X))W] + \mathbb{E}[h(X)W].$$

For the first term, condition on X . Since $\mathbb{E}[Z - e(X) \mid X] = 0$, we have $\mathbb{E}[(Z - e(X))W \mid X] = \text{Cov}(Z, W \mid X)$. For binary Z , $\text{Cov}(Z, W \mid X) = e(X)\{1 - e(X)\}\{\mathbb{E}[W \mid Z = 1, X] - \mathbb{E}[W \mid Z = 0, X]\}$. For $W = D$ this contrast is $p(X)$. For $W = Y$, Assumptions 1–3 give the conditional Wald identity $\mathbb{E}[Y \mid Z = 1, X] - \mathbb{E}[Y \mid Z = 0, X] = p(X)\text{LATE}(X)$. Taking expectations over X yields S_D and S_Y . For the second term, $\mathbb{E}[h(X)W] = \mathbb{E}[h(X)\mathbb{E}[W \mid X]] = \mathbb{E}[h(X)m_W(X)] = \text{Cov}(h(X), m_W(X))$, the last equality because $\mathbb{E}[h(X)] = 0$. This gives θ_D, θ_Y and (3). Equation (4) follows by subtracting $\bar{L} = S_Y/S_D$ from (3) and simplifying, and $S_D + \theta_D = \mathbb{E}[\tilde{Z}D]$ by construction. $\square$

Proof of Proposition 2

By Assumption 6(i), $e_t(1 - e_t) \rightarrow 0$ a.e. and in L^1 . For $S_{D,t}$, $0 \leq e_t(1 - e_t)p_t \leq \frac{1}{4}$ since $p_t \in [0, 1]$, so bounded convergence gives $S_{D,t} = \mathbb{E}[e_t(1 - e_t)p_t] \rightarrow 0$. For $S_{Y,t}$ the

integrand $e_t(1 - e_t)p_t\text{LATE}_t \rightarrow 0$ a.e., and no fixed envelope dominates it because LATE_t varies with t . We instead invoke Vitali's theorem. The family is uniformly integrable because $|e_t(1 - e_t)p_t\text{LATE}_t| \leq \frac{1}{4}|\text{LATE}_t|$ and $\text{LATE}_t \rightarrow \text{LATE}^*$ in L^2 , so $\{\text{LATE}_t\}$ is L^1 -uniformly integrable, giving $S_{Y,t} = \mathbb{E}[e_t(1 - e_t)p_t\text{LATE}_t] \rightarrow 0$. By Assumption 6(ii)–(iii), $h_t \rightarrow h^*$ and $m_{D,t}, m_{Y,t}$ converge in L^2 , so by Cauchy–Schwarz $\theta_{D,t} = \text{Cov}(h_t, m_{D,t}) \rightarrow \text{Cov}(h^*, m_D^*)$ and likewise $\theta_{Y,t} \rightarrow \text{Cov}(h^*, m_Y^*)$. The denominator of (3) converges to $\text{Cov}(h^*, m_D^*)$, nonzero by Assumption 6(iv), so (5) follows from the continuous mapping theorem. That the limit can fall outside the LATE hull or flip sign is shown by the construction in Section 6, where all $\text{LATE}(x) > 0$ yet the limit exceeds $\sup_x \text{LATE}(x)$. $\square$

Proof of Corollary 1

The population conditional first-stage slope from regressing D on $\tilde{Z}$ is $\pi_t = \mathbb{E}[\tilde{Z}_t D] / \text{Var}(\tilde{Z}_t)$. By Lemma 1, $\mathbb{E}[\tilde{Z}_t D] = S_{D,t} + \theta_{D,t} \rightarrow \text{Cov}(h^*, m_D^*)$, and $\text{Var}(\tilde{Z}_t) = \mathbb{E}[e_t(1 - e_t)] + \mathbb{E}[h_t^2] \rightarrow \mathbb{E}[(h^*)^2]$ by Assumption 6(i)–(ii). Hence $\pi_t \rightarrow \text{Cov}(h^*, m_D^*) / \mathbb{E}[(h^*)^2] \neq 0$. The conditional F has population analogue $n \pi_t^2 \text{Var}(\tilde{Z}_t) / \sigma_v^2$ with $\sigma_v^2 = \text{Var}(D - \pi_t \tilde{Z} - X'c)$ bounded, so it is bounded away from zero and diverges in n . $\square$

Proof of Proposition 3

Fix the joint law of (Z, D, X) with $h \neq 0$. Then $e(\cdot)$, $\ell(\cdot)$, $h(\cdot)$, $m_D(\cdot)$, $p(\cdot)$, $S_D = \mathbb{E}[e(1 - e)p]$, $\theta_D = \mathbb{E}[h(X)D]$ and $\text{Var}(\tilde{Z})$ are all determined, so any functional T of this law takes a fixed value. By (4), $\beta_{2\text{SLS}} - \bar{L} = (\theta_Y - \bar{L}\theta_D) / (S_D + \theta_D)$ with $\theta_Y = \mathbb{E}[h(X)Y]$ and $\bar{L} = S_Y / S_D$, $S_Y = \mathbb{E}[(Z - e(X))Y]$. Both θ_Y and S_Y are determined by the conditional law of Y given (Z, X) , which is not restricted by the marginal law of (Z, D, X) . Construct two outcome laws sharing that marginal but with conditional means m_Y and m'_Y that differ on the span orthogonal to $(1, X')$, i.e. $\text{Cov}(h, m_Y) \neq \text{Cov}(h, m'_Y)$. Because $h \neq 0$ such a perturbation exists. The perturbation changes only the level $\mathbb{E}[Y(d) \mid X]$ of the potential outcomes and leaves $D = D(Z)$, the treatment law, and the conditional LATEs untouched, so Assumptions 1–3 are preserved and both DGPs are admissible IV models. Then the two

biases differ while T is identical across them. Since the gap $\text{Cov}(h, m_Y) - \text{Cov}(h, m'_Y)$ can be made arbitrarily large (scale the orthogonal component of m'_Y), the bias is unbounded as a function of the outcome law at fixed T . Hence T carries no information about the bias. When $h \equiv 0$ the construction is empty because $\text{Cov}(h, \cdot) \equiv 0$, consistent with the first stage then certifying no contamination. $\square$

Proof of Proposition 4

$\hat{\theta}$ is a two-step estimator, a first-step OLS $\hat{\delta}$ from regressing Z on $(1, X', b(X)')$, and second-step sample means of $\hat{h}_i W_i$, $W \in \{D, Y\}$. Under the stated moment conditions and fixed K , $\hat{\delta} \xrightarrow{p} \delta$ and is asymptotically linear with influence $G^{-1}b(X_i)u_i$, $G = \mathbb{E}[bb']$. A mean-value expansion of $\hat{\theta}_W = n^{-1} \sum_i b(X_i)' \hat{\delta} W_i$ around δ gives $\sqrt{n}(\hat{\theta}_W - \theta_W) = n^{-1/2} \sum_i \{b(X_i)' \delta W_i - \theta_W\} + \mathbb{E}[Wb(X)'] \sqrt{n}(\hat{\delta} - \delta) + o_p(1)$, which is the influence function stated. The central limit theorem yields $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Sigma)$ with $\Sigma = \text{Var}(\psi_i)$. Under H_0 , $\theta = 0$ and $W_n = n\hat{\theta}'\hat{\Sigma}^{-1}\hat{\theta} \xrightarrow{d} \chi_2^2$ provided $\hat{\Sigma} \xrightarrow{p} \Sigma \succ 0$. Consistency against fixed alternatives and the local power statement are immediate from $\hat{\theta} \xrightarrow{p} \theta \neq 0$ and $\theta = \eta/\sqrt{n}$, respectively. Consistency of the pairs bootstrap for Σ follows from Hadamard differentiability of $\hat{\theta}$ in the empirical measure. $\square$

Proof of Proposition 5

Fix a sequence $\{P_n\}$ as stated. The estimator $\hat{\theta}$ and its influence function $\psi_i = (b(X_i)' \delta W_i - \theta_W)_{W \in \{D, Y\}} + (\mathbb{E}[Wb(X)'])_W G^{-1}b(X_i)u_i$ from the proof of Proposition 4 involve the first-stage strength nowhere. They are determined by the reduced-form regression of Z on $(1, X', b(X)')$ and by the cross-moments $\mathbb{E}[Wb(X)']$, none of which is the first-stage covariance $\mathbb{E}[\tilde{Z}D]$. The weak-instrument problem is a small-denominator problem for $\beta_{2SLS} = \mathbb{E}[\tilde{Z}Y]/\mathbb{E}[\tilde{Z}D]$, and that denominator does not appear. The uniform fourth-moment bound and $\Sigma_n \rightarrow \Sigma_* \succ 0$ posited for the null sequence supply a Lyapunov condition, so the Lindeberg–Feller central limit theorem for triangular arrays applies to $n^{-1/2} \sum_i \psi_{i,n}$, giving $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \Sigma_*)$ along the sequence. Under H_0 , $\theta = 0$ and $\hat{\Sigma} \xrightarrow{p} \Sigma_*$, so $W_n \xrightarrow{d} \chi_2^2$.

The generated-regressor term $\mathbb{E}[Wb(X)']G^{-1}b(X_i)u_i$ has variance of order $\text{Var}(u_i) \rightarrow 0$ as Z becomes collinear, but the leading term $b(X_i)'\delta W_i - \theta_W$ has variance bounded below because the curvature stays nondegenerate, $h_n \rightarrow h^* \neq 0$ and hence $\delta \rightarrow \delta^* \neq 0$. This delivers $\Sigma_* \succ 0$. The identical argument applied to the orthogonalized influence ψ_i^o , which is likewise free of the first-stage covariance, gives $W_n^o \xrightarrow{d} \chi_2^2$. Finally, along the contaminated sequence of Assumption 6 the same central limit theorem gives $\sqrt{n}(\hat{\theta} - \theta_n) \xrightarrow{d} N(0, \Sigma_*)$ while $\theta_n \rightarrow \theta^*$ with $\theta_D^* = \text{Cov}(h^*, m_D^*) \neq 0$; the noncentrality $n\theta_n'\Sigma_n^{-1}\theta_n \rightarrow \infty$, so $W_n \rightarrow \infty$ in probability and the power tends to one. $\square$

Proof of Proposition 6

Mean zero at the truth is immediate, because $\mathbb{E}[\psi_W^o] = \mathbb{E}[h(X)W] + \mathbb{E}[\tilde{m}_W(X)(Z - e(X))] - \theta_W = \theta_W + 0 - \theta_W = 0$, using $\mathbb{E}[Z - e(X) \mid X] = 0$. For orthogonality with respect to e , perturb $e \rightarrow e + t\Delta$. Because $h = e - L(e \mid 1, X)$ is linear in e and the linear projection is self-adjoint, $\partial_t \mathbb{E}[h(X)W] \big|_{t=0} = \mathbb{E}[(\Delta - L(\Delta \mid 1, X))m_W(X)] = \mathbb{E}[\Delta \tilde{m}_W(X)]$, while $\partial_t \mathbb{E}[\tilde{m}_W(X)(Z - e(X))] \big|_{t=0} = -\mathbb{E}[\tilde{m}_W(X)\Delta(X)]$. The two terms cancel. Perturbing $m_W \rightarrow m_W + t\Xi$ moves only the correction term, with derivative $\mathbb{E}[\tilde{\Xi}(X)(Z - e(X))] = \mathbb{E}[\tilde{\Xi}(X)\mathbb{E}(Z - e(X) \mid X)] = 0$. Hence ψ_W^o is Neyman-orthogonal in both nuisances. The affine projection $L(\cdot \mid 1, X)$ is not a third nuisance but a linear functional of e : the e -perturbation above moves $h = e - L(e \mid 1, X)$ through the projection, and the same self-adjointness that makes the two derivatives cancel annihilates its first-order contribution to the pathwise derivative. In the estimator the projection is a parametric $\sqrt{n}$ step, an OLS of Z on $(1, X)$; its estimation error contributes a regular $O_p(n^{-1/2})$ term to the influence function, not orthogonalized away but asymptotically linear, which the pairs bootstrap captures by recomputing the projection, the curvature basis, and the level functions on each resample. With K -fold cross-fitting the plug-in decomposes as $\hat{\theta}_W^o - \theta_W = n^{-1} \sum_i \psi_W^o(O_i; e, m_W) + (\text{bias}) + (\text{remainder})$. Orthogonality bounds the bias by the product $\|\hat{e} - e\|_{L^2} \|\hat{m}_W - m_W\|_{L^2} = o_p(n^{-1/2})$, and cross-fitting makes the remainder $o_p(n^{-1/2})$ under the stated L^2 rates and fourth moments (Chernozhukov et al., 2018). The leading term is a sample average of the

fixed influence ψ_i^o , so $\sqrt{n}(\hat{\theta}^o - \theta) \xrightarrow{d} N(0, \Sigma_o)$ by the central limit theorem, and $W_n^o \xrightarrow{d} \chi_2^2$ under H_0 follows as in the proof of Proposition 4. $\square$

Proof of Propositions 7 and 8

Write the numerator of (11) as $n^{-1} \sum_i (Z_i - \hat{e}(X_i)) Y_i = n^{-1} \sum_i (Z_i - e(X_i)) Y_i - n^{-1} \sum_i (\hat{e}(X_i) - e(X_i)) Y_i$. The first term converges in probability to $\mathbb{E}[(Z - e(X))Y] = S_Y$ by the argument in the proof of Lemma 1. For the second, by Cauchy-Schwarz its magnitude is at most $(n^{-1} \sum_i (\hat{e}(X_i) - e(X_i))^2)^{1/2} (n^{-1} \sum_i Y_i^2)^{1/2} = \|\hat{e} - e\|_n \cdot O_p(1)$, where $\|\hat{e} - e\|_n^2 = n^{-1} \sum_i (\hat{e}(X_i) - e(X_i))^2$ is the in-sample empirical norm. This is $o_p(1)$ whenever $\|\hat{e} - e\|_n = o_p(1)$. For the fixed, low-dimensional basis used throughout, the empirical norm converges to the population norm $\|\hat{e} - e\|_{L^2}$ by a uniform law of large numbers, so mere L^2 -consistency of the propensity estimator suffices, with no orthogonality or rate requirement. For a flexible, high-dimensional or machine-learning $\hat{e}$ trained in-sample, population L^2 -consistency need not control the empirical norm; one should then cross-fit $\hat{e}$, evaluating it on observations not used to fit it, so that the displayed average is genuinely out-of-sample and the argument goes through unchanged. The denominator converges to $\mathbb{E}[(Z - e(X))D] = S_D$ identically, and provided $S_D \neq 0$ in the fixed data-generating process the ratio converges to $S_Y/S_D = \bar{L}$, proving Proposition 7. (Asymptotic normality and inference for $\hat{\bar{L}}$, as opposed to consistency, do require controlling the first-step influence, for example a Neyman-orthogonalized moment with cross-fitting, and, because S_D may be small, weak-instrument-robust confidence sets. See Remark 1 and Section 5.) For Proposition 8, the corrected estimator's first-stage covariance is exactly this denominator $\mathbb{E}[(Z - e(X))D] = S_D = \mathbb{E}[e(X)\{1 - e(X)\}p(X)]$, which by Proposition 2 tends to 0 under Assumption 6. The identity of Corollary 2 is immediate: $\varrho = \theta_D/\mathbb{E}[\tilde{Z}D]$ by (10) and $S_D = \mathbb{E}[\tilde{Z}D] - \theta_D$, so $S_D = (1 - \varrho)\mathbb{E}[\tilde{Z}D]$; the concentration-parameter statement follows because the corrected instrument $Z - e(X)$ has covariance S_D with D and variance $\mathbb{E}[e(1 - e)]$. $\square$

Asymptotics of the bias-targeting statistic T_n^*

Let $\hat{g} = n^{-1} \sum_i \hat{h}_i(Y_i - \hat{\bar{L}} D_i)$ estimate $g(\bar{L}) = \mathbb{E}[h(X)(Y - \bar{L}D)]$, which equals $\theta_Y - \bar{L}\theta_D$ and is zero under H_0^* by (4). Expanding in $(\hat{\delta}, \hat{\bar{L}})$ about $(\delta, \bar{L})$, $\sqrt{n} \hat{g} = n^{-1/2} \sum_i \{h(X_i)(Y_i - \bar{L}D_i) - g(\bar{L})\} + \mathbb{E}[b(X)'(Y - \bar{L}D)] \sqrt{n}(\hat{\delta} - \delta) - \theta_D \sqrt{n}(\hat{\bar{L}} - \bar{L}) + o_p(1)$. We require, beyond Assumption 7 and S_D bounded away from zero, that $\hat{\bar{L}}$ be $\sqrt{n}$ -asymptotically linear with an influence function ϕ_i , $\sqrt{n}(\hat{\bar{L}} - \bar{L}) = n^{-1/2} \sum_i \phi_i + o_p(1)$. This is a genuine additional condition. Proposition 7 delivers only consistency of $\hat{\bar{L}}$, so its asymptotic linearity must be supplied separately, for instance by the Neyman-orthogonalized, cross-fitted estimator of Proposition 6 and Remark 1 under those rate conditions, which is why we invoke that construction rather than the plug-in when we need an interval. Given it, $\sqrt{n}(\hat{\delta} - \delta)$ and $\sqrt{n}(\hat{\bar{L}} - \bar{L})$ are both asymptotically linear, so $\sqrt{n} \hat{g} \xrightarrow{d} N(0, v)$ for a finite $v > 0$. In this fixed, strongly identified data-generating process the pairs bootstrap is first-order valid for the smooth functional $\hat{g}$ by the same Hadamard-differentiability argument as in Proposition 4, so the bootstrap variance $\hat{v}$ satisfies $n\hat{v} \xrightarrow{p} v$. Then $T_n^* = \hat{g}^2/\hat{v} = (\sqrt{n} \hat{g})^2/(n\hat{v}) \xrightarrow{d} \chi_1^2$ under H_0^* , and $\hat{g} \xrightarrow{p} g(\bar{L}) \neq 0$ gives consistency against any fixed alternative with nonzero bias. (Equivalently, one may write the statistic as $n\hat{g}^2/\hat{v}_1$ with $\hat{v}_1 \xrightarrow{p} v$ a consistent estimate of the asymptotic variance of $\sqrt{n} \hat{g}$; the two forms coincide since $\hat{v}_1 = n\hat{v}$.) As $S_D \rightarrow 0$ the term $-\theta_D \sqrt{n}(\hat{\bar{L}} - \bar{L})$ ceases to be $O_p(1)$, $\hat{\bar{L}}$ is no longer $\sqrt{n}$ -consistent, and both the Gaussian limit and the bootstrap approximation fail. We do not establish size control for T_n^* at that boundary. The favourable finite-sample behaviour over the collinearity range in Table B.8 is a simulation finding, not a proof of weak-signal validity, and this is exactly why the protocol pairs T_n^* with the screening test, whose size is controlled along the collinearity sequence (Proposition 5). $\square$

Appendix B. Additional Monte Carlo evidence

This appendix collects the size-and-power, power-curve, collinearity-sweep, and share-estimation exhibits summarized in Section 6. The design is that of Section 6.

The power curve sweeps the linear-probability curvature knob b for three sample sizes.

Table B.6: Directed contamination test. Panel A, one dataset, $n = 100\,000$. Panel B, rejection rates, 5% level, 300 replications, $n = 3000$, cubic basis.

<i>Panel A. Single contaminated dataset</i>					
$\hat{\theta}_D$	$\hat{\theta}_Y$	p -value	naive β_{2SLs}	cond. F	corrected $\hat{L}$
0.067	0.802	$< 10^{-4}$	8.41	20 723	1.01

<i>Panel B. Size and power</i>		
design	rejection rate (5%)	mean cond. F
SIZE (linear propensity, H_0 true)	0.013	572
POWER (curved propensity, H_0 false)	1.000	599

The top axis of Figure B.3 maps b to the induced population bias $\beta_{2SLs} - \bar{L}$. At $b = 0$ the propensity is exactly linear and the test controls size, rejecting at 3.5%, 0.5% and 1.0% for $n = 1000, 3000, 9000$ against the 5% nominal level (the $b = 0$ row of Table B.7), so the pairs bootstrap is mildly conservative rather than exactly calibrated in this design. Power then rises monotonically in b and in n , while the mean conditional F stays pinned near 1740, so the same F accompanies a bias of 0 and a bias of 2.68.

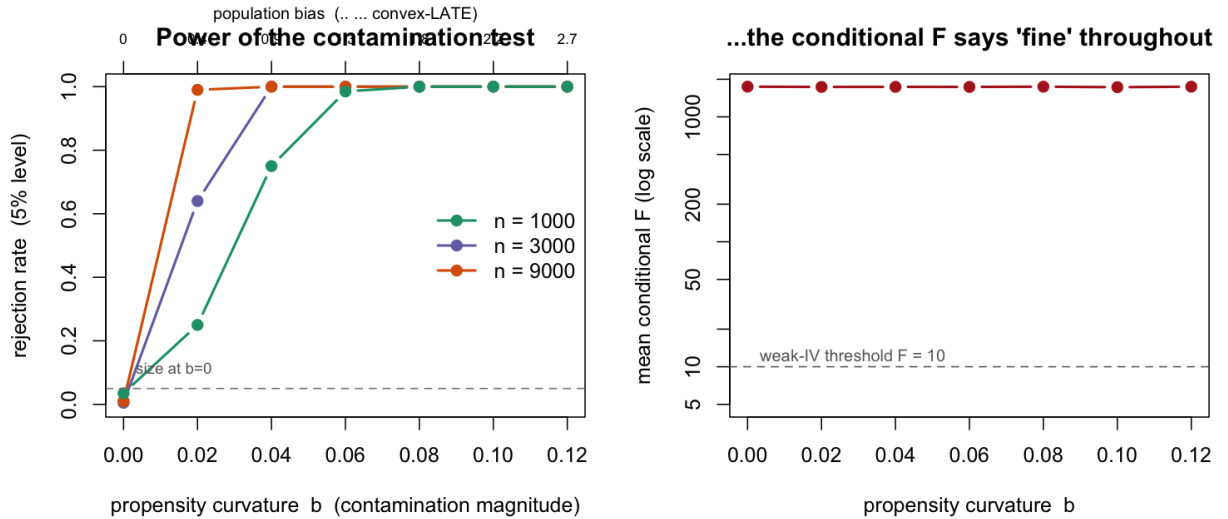


Figure B.3: Left panel. Rejection rate of the directed test vs. propensity curvature b (bottom axis) and induced population bias (top axis), for $n \in \{1000, 3000, 9000\}$, with the dashed line at the 5% level. Right panel. The mean conditional F (log scale) is invariant to b , sitting near 1740 throughout.

Table B.7: Power curve. Rejection rates (5% level, 200 replications, $B = 99$) by curvature knob b and sample size. Induced population bias $\beta_{2SLS} - \bar{L}$ at $n = 200\,000$. Mean conditional F at $n = 9000$.

b	bias	$n = 1000$	$n = 3000$	$n = 9000$	mean cond. F
0.00	−0.00	0.035	0.005	0.010	1744
0.02	0.42	0.250	0.640	0.990	1736
0.04	0.85	0.750	1.000	1.000	1738
0.06	1.33	0.985	1.000	1.000	1736
0.08	1.79	1.000	1.000	1.000	1744
0.10	2.22	1.000	1.000	1.000	1728
0.12	2.68	1.000	1.000	1.000	1743

Table B.8: Directed-S vs. Directed-X as instrument–covariate collinearity rises, in the contaminated design (rejection = power, the bias it faces in the third column) and the harmless design (rejection = size, 5% nominal). 300 replications, $n = 3000$, cubic basis. Directed-S false-alarms in every harmless row because it tests the sufficient condition $\theta = 0$. Directed-X keeps full power against genuine bias and holds size throughout, so the pairs bootstrap protects it against the weak-identification caveat of Corollary 2 over this range.

κ	$\text{Var}(Z \mid X)$	Contaminated (power)			Harmless (size)	
		bias	Directed-S	Directed-X	Directed-S	Directed-X
0.5	0.230	2.54	1.00	1.00	1.00	0.040
2.0	0.119	6.27	1.00	1.00	1.00	0.050
8.0	0.027	7.81	1.00	1.00	1.00	0.047
16.0	0.014	7.85	1.00	1.00	1.00	0.030

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Table B.9: Contamination share. Population value, plug-in estimate at $n = 3000$, 95% percentile-bootstrap coverage of the population share (250 replications, $B = 149$), and the induced 2SLS bias.

κ	ϱ (pop.)	$\hat{\varrho}$ ($n=3000$)	95% coverage	bias
0.25	0.022	0.023	0.928	1.283
0.50	0.078	0.079	0.936	2.501
1.00	0.240	0.238	0.936	4.363
2.00	0.495	0.495	0.944	6.297
4.00	0.680	0.684	0.964	7.428
8.00	0.741	0.741	0.948	7.853
16.0	0.754	0.756	0.924	7.910

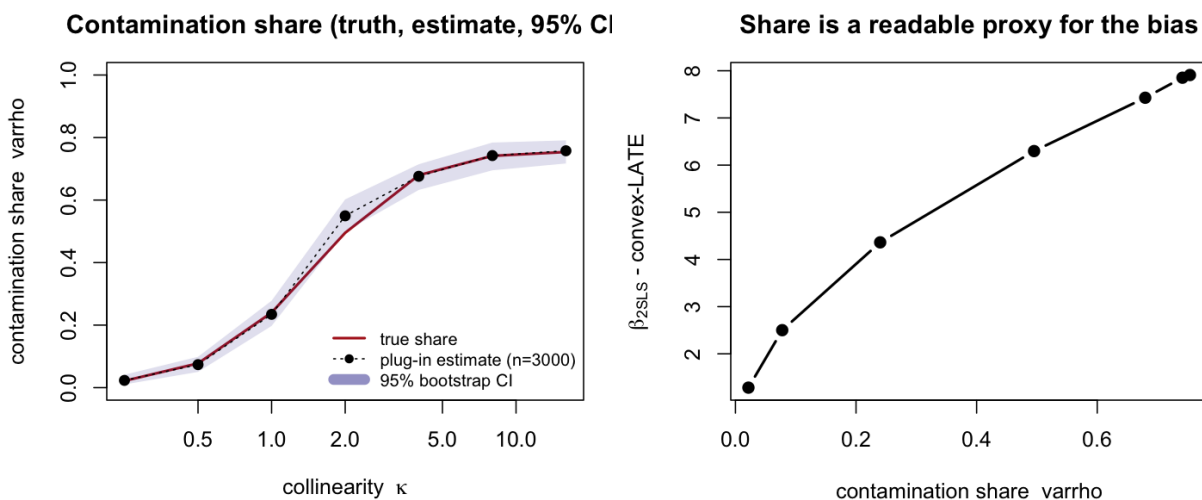


Figure B.4: Left panel. The population contamination share (solid), the plug-in estimate at $n = 3000$ (points) and its 95% bootstrap band, against the collinearity knob κ . Right panel. Along this one-parameter family the population bias is monotone in the share, so here $\hat{\varrho}$ orders these designs by danger. This ordering is a feature of the family swept, not a general property. Across unrelated designs bias also depends on θ_Y and $\bar{L}$, so equal shares can accompany different biases.

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